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CPA-Net: Physics-Assisted Attention Network for Polarimetric SAR Oil-Spill Detection

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***Abstract*—Marine oil-spill detection in polarimetric synthetic aperture radar (PolSAR) imagery remains challenging, especially when oil slicks exhibit scattering responses similar to biogenic slicks or low-wind areas. Although deep learning (DL) has improved segmentation performance, many existing models treat PolSAR data as generic multi-channel imagery and do not fully exploit complex-valued channels and scattering physics. To address this limitation, this study proposes a polarimetric attention network (CPA-Net), which integrates a physics-assisted polarimetric block (PG-CPB) into a complex-valued U-Net backbone with channel and spatial attention. PG-CPB extracts interpretable descriptors from the coherency matrix $\mathbf{T}_3$, including scattering power ratios, normalized complex coherences, and phase-based features, enabling the network to learn from both amplitude and phase information associated with oil-covered sea surfaces. Experiments, performed using the UAVSAR quad-polarimetric oil-spill detection dataset, demonstrate that CPA-Net achieves a mean intersection over union (mIoU) of 0.9556, an F1-score of 0.9755, an overall accuracy of 0.9900, and Cohen's κ of 0.9739, outperforming baseline methods. Results show that incorporating polarimetric scattering knowledge into complex-valued deep networks improves oil-spill segmentation accuracy, stability, and interpretability. This study provides a physics-assisted learning framework for operational marine PolSAR oil-spill monitoring and contributes to the integration of physical knowledge with DL for remote sensing applications.**

***Index Terms*—oil-spill detection, quad-polarimetric SAR, UAVSAR, semantic segmentation, remote sensing**

I. Introduction

MARINE oil-spills pose a significant and persistent threat to the global marine ecosystem, causing severe ecological degradation, economic losses, and long-term contamination of coastal environments [1]. Therefore, accurate and timely detection of oil-contaminated zones is crucial to minimize environmental damage and support efficient remediation strategies [2]. Oil-spills can be quickly detected and monitored through remote sensing [3], [4], due to its synoptic observation ability [5]. This enables effective response, damage assessment, and environmental impact analysis in large areas, even in remote regions. Among the available remote sensing instruments, the Synthetic Aperture Radar (SAR) has emerged as one of the most reliable technologies for marine surveillance [6], [7] due to its ability to operate in all-weather, day and night conditions and its sensitivity to variations in sea-surface roughness [8]–[11], which is particularly effective in oil-spill detection [12]–[16]. Traditional detection methods, including statistical thresholding [17], texture descriptors [18], scattering-based models [19]–[21] and machine learning (ML) classifiers [22], have been widely used for this task. However, despite these technological advancements, accurate discrimination of oil-spills from look-alike phenomena such as biogenic films, low-wind zones, and algal blooms is still a key challenge [11], [23]. These natural phenomena often exhibit similar radar backscattering in single or dual-polarimetric (DP) SAR intensity imagery, resulting in high false-alarm rates and limited generalization in operational settings [24]. Conventional linear-polarized DP SARs result in almost uncorrelated co- and cross-polarized measurements over distributed reflection-symmetry scenes. This means that the available information just relies on intensity channels. In addition, the cross-polarized channel exhibits limited sensitivity to slick-covered sea surface under low-to-moderate wind conditions. That is why DL detectors exploiting DP SAR imagery struggle to generalize across scenes [25] and typically extract the majority of the information from the co-polarized intensity channel [26].

Conversely, quad-polarimetric (QP) SAR data record both amplitude and the interchannel phase among multiple polarization channels (HH, HV, VH, VV), capturing richer scattering information and depolarization effects that can reveal subtle differences between oil-spills and natural ocean surfaces [27]. QP data have been shown to be capable of providing information about oil slick thickness and water fraction through polarimetric parameters [28]; in addition, polarimetric descriptors are also used, along with NN, to infer about oil properties [16]. However, interpreting polarimetric radar data is a complex challenge. Sea surface backscatter changes significantly with wind and waves. This makes automated detection of features like oil-spills a nontrivial task. In addition, unlocking the full potential included in QP SAR data to trigger reliable, automated oil-spill detection approaches remains a key research challenge.

In recent years, advances in DL have markedly improved the performance of SAR image interpretation and segmentation

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tasks [29]. Models such as U-Net [30], DeepLabv3+ [31], and other encoder–decoder frameworks have been successfully adapted for oil-spill segmentation, enabling hierarchical feature extraction and spatial context aggregation [32]. Nevertheless, these data-driven networks typically operate as black boxes, learning statistical correlations from pixel distributions without explicit awareness of underlying electromagnetic scattering physics. Consequently, their feature representations may be non-physical, sensor-biased, and less robust to changing environmental or imaging conditions. Moreover, ignoring phase information during training further limits their ability to exploit the full information content resulting from polarimetric SAR data [33], [34].

A complementary direction that is increasingly relevant in remote-sensing segmentation with limited labeled data is semi-supervised learning. For SAR-based scene interpretation, where pixel-level annotations are costly, semi-supervised methods can leverage large pools of unlabeled SAR imagery alongside a small annotated set. Recent approaches, such as tri-learning frameworks [35] exploit complementary views or network branches to generate reliable pseudo-labels for unlabeled samples, thereby reducing the annotation burden without sacrificing accuracy. In the context of oil-spill detection, where the quad-polarimetric oil-spill detection (QPOSD) dataset provides only 41 annotated sample images. [36] , semi-supervised strategies represent a promising avenue to extend model generalization beyond the labeled training distribution. However, semi-supervised approaches typically assume that the feature space of unlabeled samples is similar to that of labeled samples, which is not always guaranteed under diverse environmental conditions, such as different wind regimes, spillages, or acquisition geometries.

To address these gaps, physics-aware DL, which integrates domain knowledge and physical scattering models into neural network models have been recently investigated with promising results. Examples in the literature include polarimetric decomposition in remote sensing [37] and polarimetric DL models for marine spills [38]. Although these methods demonstrate improved interpretability, they typically include physical parameters as auxiliary features or regularizers, rather than embedding them directly into the network's attention or modulation mechanisms. Similarly, attention-based models such as CBAM [39] and transformer-style self-attention [40] have enhanced feature selection in remote-sensing imagery, but generally lack explicit connections to polarimetric scattering theory.

To overcome these limitations, this study proposes a physics-assisted feature extraction block (PG-CPB) that explicitly includes physical scattering characteristics into the attention generation process. The PG-CPB derives real-valued feature maps directly from the coherency matrix T, incorporating parameters such as co-polarization ratios, cross-polarization ratios, and normalized complex coherences. These features capture the intrinsic amplitude and phase relationships that can be linked to the underlying scattering process, providing physically interpretable information about the damping and the depolarization associated to the slick-covered sea surface and providing, therefore, a richer feature space to discriminate between oil and look-alikes. The PG-CPB then uses these physically derived features to generate both channel and spatial attention maps, which are applied to modulate the complex-valued feature tensors within the network. This design allows the model to retain the amplitude–phase consistency of polarimetric features while emphasizing physically meaningful scattering patterns.

Therefore, a complex polarimetric attention network (CPA-Net) is developed by embedding PG-CPB modules within a complex-valued U-Net backbone. The resulting model combines the power of deep neural networks with the interpretability of polarimetric scattering features. It not only enhances detection accuracy, but also provides physically meaningful attention responses, improving transparency and robustness against look-alike phenomena. To validate the framework, the UAVSAR quad-polarimetric oil-spill detection (QPOSD) dataset is employed [41]. This dataset includes a variety of scenes captured under diverse environmental conditions, allowing the model's robustness and generalization ability to be tested in real-world scenarios. Experimental results demonstrate that CPA-Net consistently outperforms conventional deep networks and achieves stronger generalization, while maintaining clear physical interpretability of learned attention maps.

The main contributions of this work can be summarized as follows.

- A novel PG-CPB that fuses power ratios, polarimetric coherences, and phase characteristics derived from the coherency matrix T is proposed.
- A physics-assisted polarimetric attention net, namely the CPA-Net, that preserves amplitude–phase relationships during learning is proposed.
- An extensive experimental evaluation on QP SAR dataset for multi-class oil-spill segmentation is proposed.
- The improved interpretability, physical consistency, and robustness against look-alike phenomena is discussed against state-of-the-art methods.

By embedding explicit polarimetric physics into attention computation, this study advances the integration of physical scattering theory and deep neural modeling, providing a transparent, interpretable, and effective framework for understanding complex-valued SAR images. The remainder of the manuscript is organized as follows: section II provides physical background. The PG-CPB and the network architecture are described in section III and section IV, respectively. Section V provides information about the dataset and the experimental setup, while section VI is to discuss the experimental results. Section VII contains comments on all aspects of the work, while conclusions are drawn in section VIII.

II. Physical Background

A. Fundamentals of SAR Polarimetry

For each resolution cell, a QP SAR measures the Sinclair matrix [42], [43]:

$$\mathbf{S} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \tag{1}$$

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where S_{pq} $(p, q \in H, V)$ denotes the complex scattering amplitude for transmitted polarization p and received polarization q. Under the monostatic radar configuration and assuming the Backscattering Alignment Convention (BSA), due to the reciprocity theorem ($S_{HV} = S_{VH}$), a Pauli target vector can be associated to the Sinclair matrix [44]:

$$\mathbf{k} = \frac{1}{2}\left[S_{HH} + S_{VV},\ S_{HH} - S_{VV},\ 2S_{HV}\right]^T \tag{2}$$

The coherency matrix $\mathbf{T}$ is then formed as the outer product of the Pauli target vector [44]:

$$\mathbf{T}_3 = \langle \mathbf{k}\mathbf{k}^\dagger \rangle \tag{3}$$

where $\dagger$ represents the Hermitian conjugate transpose and $\langle\cdot\rangle$ denotes spatial averaging over a local estimation window.

$\mathbf{T}$ is a 3×3 Hermitian positive semidefinite matrix that comprehensively characterizes the scattering properties of the observed scene [44]:

$$\mathbf{T}_3 = \frac{1}{2}\begin{pmatrix} T_{11} & T_{12} & T_{13} \\ T_{12}^* & T_{22} & T_{23} \\ T_{13}^* & T_{23}^* & T_{33} \end{pmatrix} \tag{4}$$

The diagonal elements are related to three fundamental scattering mechanisms resulting from the Pauli decomposition:

- $T_{11} = \frac{1}{2}\langle|S_{HH}+S_{VV}|^2\rangle$: describes the surface scattering mechanism component.
- $T_{22} = \frac{1}{2}\langle|S_{HH} - S_{VV}|^2\rangle$: describes the dihedral or double-bounce scattering mechanism component.
- $T_{33} = 2\langle|S_{HV}|^2\rangle$: describes the volume scattering mechanism component.

While the diagonal elements describe the intensity of specific scattering mechanisms, the off-diagonal terms represent the complex correlations and phase relationships between them:

- T_{12} represents the correlation between the sum and the difference of the co-polarised returns.
- T_{13} represents the correlation between the sum of co-polarized channels and the cross-polarised return.
- T_{23} represents the correlation between the difference of co-polarized channels and the cross-polarised return.

For naturally distributed targets, according to the reflection symmetry, inter-channels correlations, i.e (T_{13}, T_{23}), should be theoretically zero. Non-zero values typically indicate tilted surfaces that induce shifts in the orientation of polarization. T_{12} represents the correlation between surface scattering and double-bounce scattering: it is often used to discriminate among man-made structures and random natural scatterers [45].

B. Theoretical Background: oil-spill signatures in SAR images

The radar backscattering behavior of the ocean surface is mainly controlled by short gravity-capillary waves, which produce resonant Bragg scattering in the microwave region [46]. For oil-free sea surfaces, the HH and VV backscattered signals are dominated by the Bragg scattering mechanism under low-to-moderate wind conditions and intermediate incidence angles. This results in relatively high values of σ^0_{HH} and σ^0_{VV}, with $\sigma^0_{\mathrm{VV}} > \sigma^0_{\mathrm{HH}}$, while σ^0_{VH} remains negligible (mostly below the system noise level). The co-polarized channels also exhibit strong correlation, resulting in high HH-VV coherence $|\gamma_{HH,VV}|$. For oil-covered sea surfaces, the viscosity and damping effects of oil suppress small-scale waves, reducing surface roughness [46]. This mechanism directly leads to a significant reduction in co-polarized backscattering coefficients $\sigma^0_{\text{co-pol}}$, a phenomenon consistently observed across L- (1–2 GHz), C- (4–8 GHz), and X-band (8–12 GHz) SAR systems [28], [47], [48]. Consequently, oil slicks appear as dark and homogeneous patches in SAR imagery.

In oil-free seas, the HH-VV coherence $|\gamma_{HH,VV}|$ is typically high due to dominant Bragg scattering; conversely, oil slicks result in lower coherence magnitude and a very fluctuating coherence phase [48]. Over oil-covered areas, while co-polarized returns (σ^0_{HH} and σ^0_{VV}) often decrease by approximately 7–15 dB due to the strong suppression of Bragg waves, the cross-polarized channel (σ^0_{HV}) generally shows a much weaker attenuation of about 1–3 dB, and in some cases may even display a slight increase depending on local sea and wind conditions [49]. Furthermore, cross-polarized scattering (HV/VH), which is typically associated with depolarization, exhibits a lower damping behavior in the presence of surface oil-spills [28]. A change in $|\gamma_{HH,VV}|$ has been also observed in [49]. Commonly used indicators to observe oil-spilled areas in multi-polarization SAR imagery include the ratios [2], [28], [47]:

$$\frac{\sigma^0_{\mathrm{HV}}}{\sigma^0_{\mathrm{VV}}}, \qquad \frac{\sigma^0_{\mathrm{HV}}}{\sigma^0_{\mathrm{HH}}}, \qquad \frac{\sigma^0_{\mathrm{VV}}}{\sigma^0_{\mathrm{HH}}}.$$

For instance, the cross-to-co-polar ratio $\frac{\sigma^0_{\mathrm{HV}}}{\sigma^0_{\mathrm{VV}}}$ typically increases to about 0.1-0.3 in oil slicks, whereas it remains below 0.05 under oil-free sea conditions [28]. Moreover, oil spills in SAR imagery create regions with more homogeneous scattering values (e.g., lower texture variance, entropy, or roughness [50]) compared to look-alikes (e.g., biogenic slicks, low-wind areas, rain cells), which often exhibit scattered, discontinuous, or heterogeneous textures [50]. This homogeneity arises from oil's uniform damping of capillary waves, resulting in a smoother backscattering patterns, while look-alikes may arise from varied environmental causes resulting in a more inhomogeneous darker area [50].

These physical distinctions are of key importance because many look-alike phenomena, including low-wind areas, biogenic slicks, upwelling zones, and internal waves, can mimic the radiometric darkening caused by oil spill, but do not reproduce the same polarimetric behavior [16]. In particular, look-alikes often maintain higher coherence, more stable phase signatures, or different cross-to-co-polarization ratios compared to true oil slicks [20]. Therefore, utilizing the full polarimetric information contained in $\mathbf{T}$ is essential to achieve robust oil–look-alike discrimination. Oil-spills exhibit a unique combination of (i) reduced co-pol power due to Bragg scattering waves, (ii) distinct cross-pol behavior, and (iii) decorrelated polarimetric signatures [51]. These properties justify the use of physics-assisted, polarimetry-aware learning strategies to leverage SAR-based oil-spill detection.

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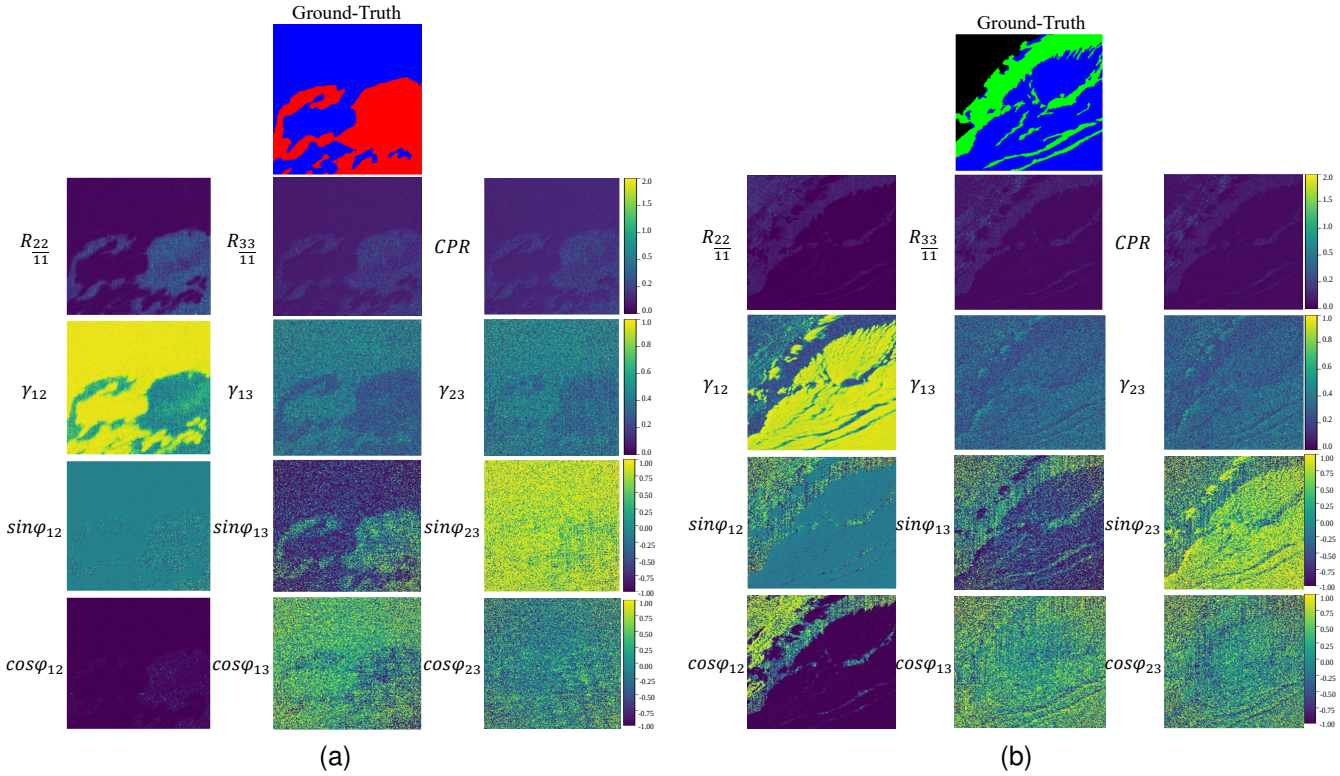


Fig. 1: Visualization of the proposed PG-CPB responses for some representative scenes. (a) oil spill scene in the open water, (b) look-alike scene, representing a coastal land in the upper left corner with the sea surface in the bottom. For each scene, each column shows the corresponding physical polarimetric features, with the power ratios, magnitude coherence and phase components shown in the first, second, third and fourth row, respectively

C. Physics-Assisted Feature Set

This study uses twelve physically interpretable features derived from the coherency matrix $\mathbf{T}$, which can be classified into three types: power-based ratios, normalized complex coherences, and phase components.

a) Power-Based Ratios: exploit the diagonal terms of $\mathbf{T}$ matrix to capture the relative dominance of different scattering mechanisms:

- $R_{33/11}$ characterizes the dominance of volume versus surface scattering, sensitive to depolarizing effects and structural complexity [52].
- $R_{22/11}$: characterizes the dominance of double-bounce versus surface scattering [53].
- Circular polarization ratio (CPR), namely the cross-pol to total co-pol power ratio, is to distinguish depolarizing targets from non-depolarizing ones [53].

$$R_{33/11} = \frac{T_{33}}{T_{11} + \varepsilon} \tag{5}$$

$$R_{22/11} = \frac{T_{22}}{T_{11} + \varepsilon} \tag{6}$$

$$\text{CPR} = \frac{T_{33}}{T_{11} + T_{22} + \varepsilon} \tag{7}$$

where $\varepsilon = 1 \times 10^{-6}$ is a small stabilizing constant chosen to be negligible relative to typical T_{11} values in L-band UAVSAR data, i.e., $\mathcal{O}(10^{-2}$–$10^{-1})$, while preventing division-by-zero errors or numerically unstable gradients in near-zero backscatter regions, such as shadowed targets or specular reflectors. In oil-spill covered areas, the damping of capillary waves by surface films strongly suppresses the surface scattering component (T_{11}) [47]. This results in an increased values of $R_{33/11}$ and CPR, while the ratio $R_{22/11}$ becomes unstable due to the significant reduction of both T_{11} and T_{22}.

b) Normalized Complex Coherences: they are to exploit the off-diagonal terms of $\mathbf{T}$, to capture the statistical dependencies among different polarimetric channels. T_{13}, T_{23}, T_{12} are complex elements, with the magnitude and phase representing the complex inter-channel correlation.
Normalized complex coefficients, ranging from 0 to 1, are defined as the normalization of the off-diagonal elements:

$$\gamma_{12} = \frac{T_{12}}{\sqrt{T_{11}T_{22}} + \varepsilon} \tag{8}$$

$$\gamma_{13} = \frac{T_{13}}{\sqrt{T_{11}T_{33}} + \varepsilon} \tag{9}$$

$$\gamma_{23} = \frac{T_{23}}{\sqrt{T_{22}T_{33}} + \varepsilon} \tag{10}$$

where $\gamma_{pq} \in \mathbb{C}$ represents the normalized complex coherence between scattering mechanisms p and q.

- γ_{12}: deals with the contributions of the co-polar channels of the $\mathbb{T}$ matrix.
- γ_{13} and γ_{23}: deal with the contributions of the co-polar and cross-polar channels of the $\mathbb{T}$ matrix.

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In an oil-free environment, the Bragg mechanism dominates the surface scattering with the co-polar contributions being very correlated, while the cross- polar contribution is typically negligible. The presence of oil dampens the capillary waves, thus resulting in changes in roughness. While the scattering mechanism remains primarily surface-based, oil can induce minor asymmetries in the polarimetric signal. In particular, $\gamma_{12} \approx 1$ in calm or moderate seas with no asymmetric structures. A lower value of γ_{12} indicates the possible presence of oil that breaks the symmetry of the surface.

c) Phase Components: To fully exploit the information in the complex coherences, each γ_{pq} is decomposed into two distinct components:

$$\sin(\phi_{pq}), \quad \cos(\phi_{pq}) \tag{11}$$

where $\phi_{pq} = \angle\gamma_{pq}$ denotes the phase of the complex coherence.

- The sine component $\sin(\phi_{pq})$ captures the odd-symmetric part of the phase relationship, sensitive to asymmetric scattering processes.
- The cosine component $\cos(\phi_{pq})$ captures the even-symmetric part of the phase relationship.

For oil-spill detection, the phase components of $|\gamma_{12}|$ and $|\gamma_{13}|$ provide the most discriminative information, carrying information about the stability of the dominant scattering mechanism. This roots in the fact that oil-spills create an area with a more homogeneous scattering values than the look-alikes ones [54].

d) Complete Feature Vector: To encapsulate the full polarimetric scattering behavior each pixel is characterized by a comprehensive 12-dimensional feature vector $\mathbf{f} \in \mathbb{R}^{12}$:

$$f = [R_{33/11},\ R_{22/11},\ \text{CPR},\ |\gamma_{13}|,\ |\gamma_{12}|,\ |\gamma_{23}|,\ \sin(\angle\gamma_{13}),$$
$$\cos(\angle\gamma_{13}),\ \sin(\angle\gamma_{12}),\ \cos(\angle\gamma_{12}),\ \sin(\angle\gamma_{23}),\ \cos(\angle\gamma_{23})]\,.$$

This physics-assisted feature set provides a complete representation of the scattering physics, enabling robust discrimination between mineral oil-spills and biogenic look-alikes based on their own polarimetric signatures. Fig.1 shows two representative examples of an oil spill (a) and a look-alike (b). The first row contains ground truth, where oil spills, look-alikes, sea surface, and land are depicted in red, green, blue, and black, respectively. The power ratios, magnitude coherence, sine component, and cosine component are shown in the second, third, fourth, and fifth rows, respectively. When dealing with oil discrimination against the surrounding water (see panel (a)), all the features show good performance with the exception of . When dealing with look-alikes discrimination against the surrounding water, most of the features do not discriminate the look-alikes from the surrounding environment. Hence, the following features γ_{13} and γ_{23} allow for better observing oil slicks and discriminating them from look-alikes. It is also worth noting that phase-related features (i.e, the last two rows in Fig.1) exhibit a very noisy behavior with the $\sin(\phi 13)$ showing ability to both detect oil spills and discriminating them form look-alikes.

The power ratio ($R_{22/11}$) emphasizes well oil with respect to the sea while de-emphasizes the look-alikes. This means that this term shows ability to distinguish oil from look-alikes. The magnitude coherence (γ_{12}) reveals a spatial pattern that clearly distinguishes oil from both water and look-alikes. The same applies to γ_{13} and γ_{23} that demonstrate a different sensitivity to oil spill and look-alikes.

D. Colorbar Standardization and Visualization

(1) Power-ratio features: The power-ratio maps $R_{33/11}$, $R_{22/11}$, and CPR are non-negative and can exhibit heavy-tailed distributions due to scene heterogeneity (e.g., rough sea states, coastal boundaries, or land contamination). Therefore, for visualization purposes, a fixed linear range based on the empirical distribution of the training set is applied, and clip only extreme outliers: *(i)* $R_{33/11}, R_{22/11}$ and CPR are visualized in $[0,\ 2]$.

(2) Coherence magnitudes: The coherence magnitudes $|\gamma_{13}|$, $|\gamma_{12}|$, and $|\gamma_{23}|$ are theoretically bounded in $[0, 1]$. Hence, all coherence magnitude maps are displayed with a fixed colorbar range $[0, 1]$ across all scenes. This choice is critical because image-wise autoscaling can artificially exaggerate or suppress coherence differences and destroy physical meaning.

(3) Phase-derived features: Phase is visualized via $\sin(\angle\gamma_{pq})$ and $\cos(\angle\gamma_{pq})$, which are bounded in $[-1, 1]$ and avoid phase-wrapping discontinuities. Accordingly, all phase-derived maps are displayed with a fixed diverging range $[-1, 1]$ centered at zero. Since phase becomes unreliable when $|\gamma_{pq}| \to 0$, phase-derived maps should be interpreted jointly with the corresponding magnitude maps; regions of very low coherence are expected to exhibit phase randomness.

III. Physics-Assisted Polarimetric Block

A. Overview

The proposed PG-CPB integrates explicit polarimetric scattering physics into the attention mechanism of a complex-valued network. This design is motivated by the physical principle that oil-spills alter both the amplitude ratios and phase coherence among polarization channels, which are quantitatively captured by the coherency matrix $\mathbf{T}$ [2]. Conventional deep learning approaches, while effective, often ignore these physically meaningful relationships and learn non-physical correlations. The PG-CPB addresses this gap through a three-stage model:

1) *Physical features extraction*: compute physically interpretable real-valued ratios and coherences from the $\mathbf{T}$ matrix.
2) *Attention generation*: derive channel and spatial attention weights from these polarimetric features.
3) *Complex modulation*: apply the learned weights to complex feature maps while maintaining amplitude-phase consistency.

The overall processing flow can be summarized as:

$$\mathbf{T} \to f \to \{\mathbf{A_c}, \mathbf{A_s}\} \to \tilde{\mathbf{X}} \to \mathbf{Y} \tag{12}$$

where f denotes the physical features tensor, $\mathbf{A_c}$ and $\mathbf{A_s}$ represent the channel and spatial attention maps, $\tilde{\mathbf{X}}$ represents

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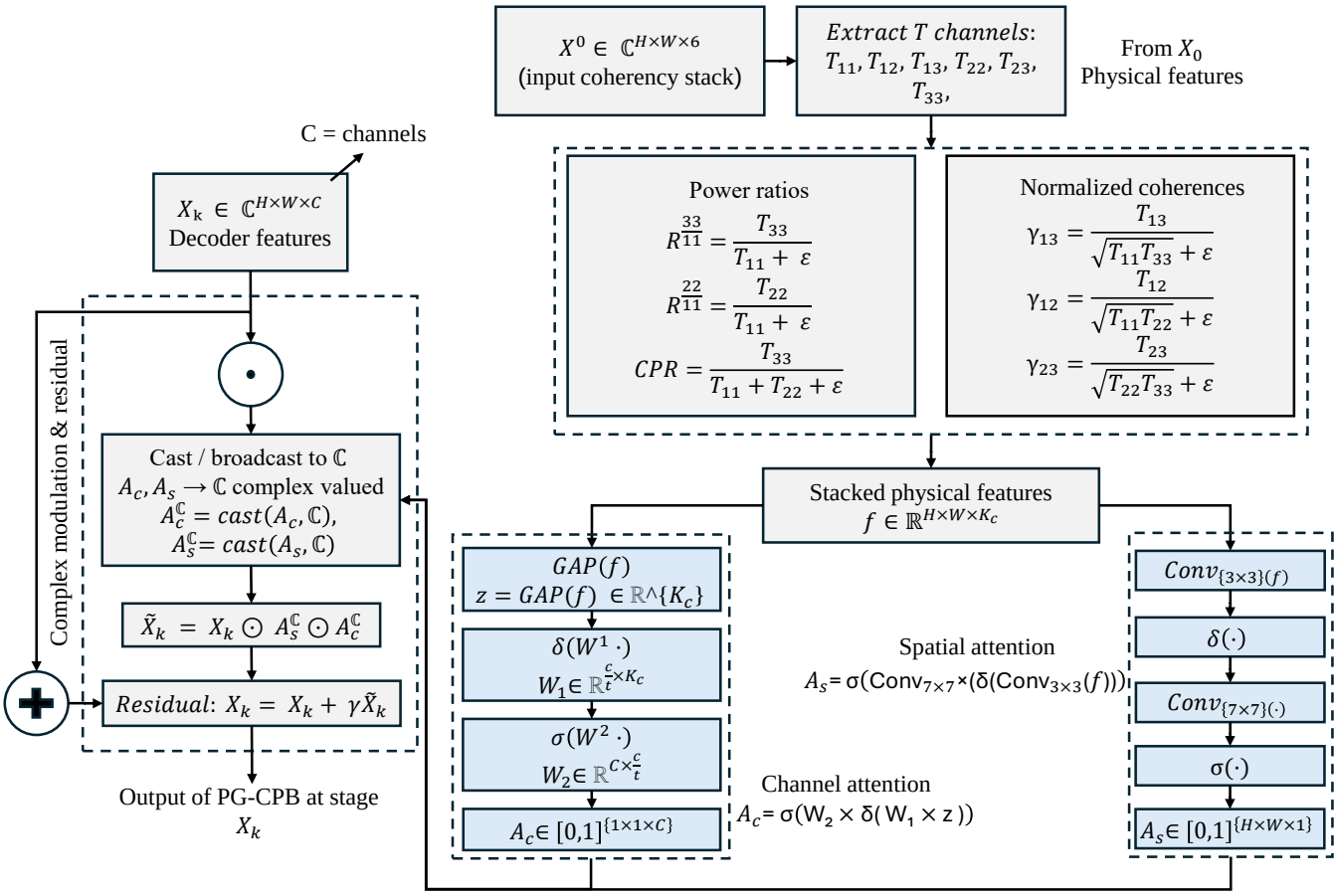


Fig. 2: PG-CPB schematic. Polarimetric features derived from the input coherency matrix are converted into channel and spatial attention maps. These attention gates are broadcast to the complex domain, used to modulate decoder features, and then merged through a learnable residual connection.

the modulated complex features, and $\mathbf{Y}$ is the residual output. The complete framework of the PG-CPB is illustrated in Fig. 2.

B. Attention Generation from Physical Features

Two complementary attention mechanisms are derived from the physical features tensor $f \in \mathbb{R}^{H\times W\times C}$.

1) Physical Channel Attention: To capture the global statistical properties of the polarimetric scattering, the spatial information of the physical features f are firstly aggregated through global average pooling (GAP) to produce a channel-wise descriptor $\mathbf{z} \in \mathbb{R}^{1\times1\times C}$:

$$\mathbf{z} = \mathrm{GAP}(f) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} f(i,j,:) \tag{13}$$

Then, a compact multi-layer perceptron (MLP) transforms this descriptor to generate the channel attention weights $\mathbf{A_c}$:

$$\mathbf{A_c} = \sigma(\mathbf{W}_2\,\delta(\mathbf{W}_1\,\mathbf{z})) \tag{14}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C/r\times C}$, $\mathbf{W}_2 \in \mathbb{R}^{C\times C/r}$ are trainable weight matrices with reduction ratio $r = 2$, $\delta(\cdot)$ denotes the ReLU activation, and $\sigma(\cdot)$ is the sigmoid function. The resulting map $\mathbf{A_c} \in [0,1]^{1\times1\times C}$ reweights each feature channel based on its physical relevance to oil-spill scattering mechanisms.

2) Physical Spatial Attention: To capture spatially localized scattering phenomena, a spatial attention map $\mathbf{A_s} \in [0,1]^{H\times W\times1}$ is generated directly from the physical features tensor *f*. This is achieved by applying a convolutional network that learns to emphasize regions with characteristic oil slick signatures:

$$\mathbf{A_s} = \sigma(\mathrm{Conv}_{7\times7}(\delta(\mathrm{Conv}_{3\times3}(f)))) \tag{15}$$

where $\mathrm{Conv}_{k\times k}$ denotes a 2D convolution with a $k\times k$ kernel. The map $\mathbf{A_s}$ preferentially activates image regions exhibiting oil-induced damping and phase decorrelation [38].

C. Complex Feature Modulation

Given a complex-valued feature tensor $\mathbf{U_k} \in \mathbb{C}^{H\times W\times C}$ from the preceding layer in decoder, the PG-CPB modulates it as follows:

$$\tilde{\mathbf{X}} = \mathbf{U_k} \odot \mathbf{A_s} \odot \mathbf{A_c}, \qquad \mathbf{U_k} = \mathbf{U_k} + \gamma\,\tilde{\mathbf{X}} \tag{16}$$

where $\odot$ represents element-wise multiplication and γ is a learnable scalar parameter initialized to zero for training stability. The attention maps $\mathbf{A_c}$ and $\mathbf{A_s}$ are real-valued but are broadcast to complex tensors before multiplication, preserving both real and imaginary components:

$$\tilde{\mathbf{X}} = \mathbf{U_k} \odot \mathrm{cast}(\mathbf{A_s}, \mathbb{C}) \odot \mathrm{cast}(\mathbf{A_c}, \mathbb{C})$$

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This design preserves the integrity of both phase and amplitude information and enhances physically meaningful scattering responses relevant to oil-spill detection.

D. Physical Interpretability

The PG-CPB maintains strong physical interpretability by directly linking each component of the feature vector f to well-established electromagnetic scattering mechanisms relevant to oil-spill detection:

- Power ratios ($R_{33/11}, R_{22/11}, \mathrm{CPR}$): These ratios capture the relative dominance of different scattering mechanisms. Oil-spills typically exhibit:
  - Increased $R_{22/11}$ and CPR: oil-spills suppress Bragg scattering from capillary waves (reducing T_{11}) more effectively than the non-Bragg scattering mechanisms that contribute to cross-polarized returns (increasing T_{33}), leading to higher ratios in oil-covered areas.
  - Unstable $R_{33/11}$: this ratio becomes unreliable over oil slicks as both T_{11} (surface scattering) and T_{33} (volumetric scattering) are strongly suppressed, resulting in numerically unstable quotients of small values.
- Coherence magnitudes ($|\gamma_{13}|, |\gamma_{12}|, |\gamma_{23}|$): these quantify the statistical correlation between different scattering mechanisms:
  - Increased $|\gamma_{12}|$ in oil slicks: oil creates a more homogeneous scattering environment where the residual surface (T_{11}) and dihedral (T_{22}) scattering components become more correlated, unlike the uncorrelated mixture of Bragg scattering and wave breaking in clean sea.
  - Low $|\gamma_{13}|$ and $|\gamma_{23}|$: these remain low over both oil-free and oil-covered sea due to the absence of significant volumetric scattering mechanisms (T_{33}) in maritime environments.
- Phase components ($\sin\phi_{pq}, \cos\phi_{pq}$): these capture the relative phase relationships between polarimetric scattering channels:
  - Relatively stable phase in slick-free sea surfaces: under suitable low-to-moderate wind conditions, Bragg scattering can produce stronger inter-channel correlation and relatively narrow phase-difference distributions.
  - Increased phase variability in oil-covered surfaces: oil-induced damping of short Bragg-scale waves alters the dominant scattering contribution and can reduce inter-channel correlation, resulting in broader phase-difference distributions [20], [28], [48], [55].

The attention mechanisms $\mathbf{A_c}$ and $\mathbf{A_s}$ naturally amplify areas exhibiting this heterogeneous characteristic "oil-spill signature" that calls for suppressed co-polarized power, enhanced cross-polarized ratios, and stabilized phase coherences. Empirical analysis demonstrates that $\mathbf{A_s}$ activates more strongly at slick boundaries where damping gradients are maximal, while $\mathbf{A_c}$ learns to suppress feature channels corresponding to scattering mechanisms that are not correlated with oil-spill physics.

E. Physical Feature Stabilization and Attention-Map Visualization

To ensure numerical stability in the computation of the polarimetric descriptors and to facilitate interpretation of the PG-CPB, two auxiliary procedures are employed, as summarized in Algorithms 1 and 2. The first procedure computes the physical descriptors directly from the coherency matrix while incorporating numerical stabilization and phase-aware representation. The second extracts the physical descriptor maps and attention responses for visualization and interpretation. These procedures do not introduce additional physical constraints into the optimization objective.

Algorithm 1 Physical Feature Computation and Numerical Stabilization for PG-CPB

Require: Coherency matrix $\mathbf{T}$; numerical stability constant $\varepsilon = 10^{-6}$.

Ensure: Real-valued physical descriptor tensor $\mathbf{f}$.

1: Extract the diagonal terms T_{11}, T_{22}, T_{33} and complex off-diagonal terms T_{12}, T_{13}, T_{23} from $\mathbf{T}$.
2: Compute the three power-ratio descriptors $R_{33/11}$, $R_{22/11}$, and CPR using ε in the denominators for numerical stability.
3: Compute the normalized complex coherence coefficients γ_{13}, γ_{12}, and γ_{23}.
4: Extract the coherence magnitudes $|\gamma_{13}|$, $|\gamma_{12}|$, and $|\gamma_{23}|$.
5: Compute the phase angles $\phi_{pq} = \angle\gamma_{pq}$.
6: Represent each phase using $(\sin\phi_{pq}, \cos\phi_{pq})$ to avoid the discontinuity at $\pm\pi$.
7: Stack the three power-ratio, three coherence-magnitude, and six phase descriptors to form $\mathbf{f} \in \mathbb{R}^{H\times W\times 12}$.
8: **return** $\mathbf{f}$

1) Physical Feature Computation and Numerical Stabilization (Algorithm 1): The physical descriptors are computed directly from the coherency matrix and consist of three power-ratio descriptors $(R_{33/11}, R_{22/11}, \mathrm{CPR})$, three coherence magnitudes $(|\gamma_{13}|, |\gamma_{12}|, |\gamma_{23}|)$, and six phase components $(\sin\phi_{13}, \cos\phi_{13}, \sin\phi_{12}, \cos\phi_{12}, \sin\phi_{23}, \cos\phi_{23})$. A small constant $\varepsilon = 10^{-6}$ is introduced in the denominator of ratio and coherence calculations to prevent numerical instability when the corresponding scattering power approaches zero. In addition, the phase of each complex coherence coefficient is represented using its sine and cosine components rather than the raw phase angle. This representation preserves the periodic nature of phase while avoiding the discontinuity between $-\pi$ and π. No Winsorization or percentile-based clipping is applied to the physical descriptors.

2) Export and Visualization of Attention Maps and Physical Features (Algorithm 2): To investigate how the physically derived descriptors guide feature modulation within PG-CPB, Algorithm 2 extracts the channel attention weights A_c, spatial attention map A_s, and the corresponding physical descriptor maps during inference. These outputs allow the attention

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responses to be compared with the underlying polarimetric characteristics of the scene.

Algorithm 2 Export and Visualization of Attention Maps and Physical Features

Require: Input coherency matrix $\mathbf{T}$; intermediate complex feature tensor $\mathbf{X}_k$; trained PG-CPB.

Ensure: Channel attention A_c, spatial attention A_s, and physical descriptor maps $\mathbf{f}$.

1: Resize the real and imaginary components of $\mathbf{T}$ to the spatial resolution of $\mathbf{X}_k$ when required.
2: Compute the 12 physical descriptors $\mathbf{f}$ using Algorithm 1.
3: Apply global average pooling to $\mathbf{f}$ and pass the resulting descriptor vector through the channel-attention layers to obtain $A_c \in [0,1]^{1\times1\times C}$.
4: Pass $\mathbf{f}$ through the spatial-attention layers to obtain $A_s \in [0,1]^{H_k\times W_k\times 1}$.
5: Store A_c, A_s, and $\mathbf{f}$ from the selected PG-CPB layer during inference.
6: If required for visualization, resize A_s and the physical descriptor maps to the input image resolution using bilinear interpolation.
7: Apply display normalization to individual maps only for visualization.
8: **return** $A_c, A_s, \mathbf{f}$

The exported physical descriptor maps provide a direct means of comparing physically interpretable polarimetric quantities, such as $|\gamma_{12}|$, $\sin\phi_{13}$, and CPR, with the learned attention responses. The spatial attention map can be resized to the input resolution and overlaid on the corresponding SAR representation, while the channel attention coefficients can be visualized as channel-wise weights. The descriptor maps may likewise be resized and normalized for display. These visualization procedures are performed after inference and therefore do not modify the inputs or affect the segmentation predictions. They provide qualitative support for interpreting how the physically derived descriptors influence the attention mechanism across oil spills, look-alikes, sea surface, and land.

F. Feature Normalization and Attention–Map Export

To ensure numerical stability and physically meaningful learning within the PG-CPB, two auxiliary procedures are implemented, exhibited in Algorithms 1 and 2. These steps do not alter the network architecture, but they standardize the physical features and provide interpretability outputs essential for scientific analysis.

IV. Network Architecture: Complex Polarimetric Attention Network

A. Overview

The proposed CPA-Net integrates complex-valued convolutional feature extraction with physics-assisted modulation derived from the coherency matrix $\mathbf{T}$. The network adopts a U-Net-style encoder-decoder model, enabling hierarchical feature representation while preserving fine spatial details through skip connections, as shown in Fig. 3.

The encoder extracts multi-scale complex-valued features; then, the bottleneck captures global contextual dependencies, and the decoder restores spatial resolution while injecting physically interpretable attention using the PG-CPB.

The end-to-end mapping learned by CPA-Net is:

$$\mathcal{F}_\theta : \mathbb{C}^{H\times W\times C_{\text{in}}} \longrightarrow \mathbb{R}^{H\times W\times K} \tag{17}$$

where $C_{\text{in}} = 6$ corresponds to the independent elements of $\mathbf{T}$ (T_{11}, T_{12}, T_{13}, T_{22}, T_{23}, T_{33}), and $K = 4$ is the number of segmentation classes (sea surface, oil-spill, look-alike, and land). The network outputs per-pixel class probabilities $\widehat{\mathbf{Y}} \in [0,1]^{H\times W\times K}$.

B. Encoder: Complex Feature Extraction

The encoder progressively captures scattering signatures across scales through stacked complex convolutional layers. For each stage $k \in \{1,2,3,4\}$, two complex convolutional layers are applied, followed by complex dropout and complex max pooling:

$$\mathbf{E}_k = \phi_{\text{ReLU}}\left(\mathbf{W}_{k,2} * \left(\phi_{\text{ReLU}}(\mathbf{W}_{k,1} * \mathbf{P}_{k-1})\right)\right) \tag{18}$$

$$\mathbf{P}_k = \mathcal{P}(\mathbf{E}_k) \tag{19}$$

where $*$ denotes complex convolution, $\phi_{\text{ReLU}}(\cdot)$ applies ReLU activation to real and imaginary components separately, and $\mathcal{P}(\cdot)$ performs complex max pooling. Specifically, we employ Cartesian ReLU (CReLU), defined as

$$\phi_{\text{CReLU}}(z) = \text{ReLU}(\Re(z)) + j\,\text{ReLU}(\Im(z)),$$

which processes real and imaginary components independently using the standard ReLU operation. This choice ensures compatibility with the CVNN framework enabling element-wise gradient computation for complex-valued features. CReLU is not strictly phase-preserving, because the phase of activations may change when either the real or imaginary component is negative and subsequently truncated. In contrast, modulus ReLU, which thresholds the activation magnitude while preserving the phase angle of retained activations, is expected to provide a much robust phase-preservation guarantees and therefore represents a useful direction for future investigation. In practice, the physics-assisted attention mechanism in PG-CPB helps mitigating potential phase distortions in intermediate layers by explicitly re-grounding the attention signal in physical phase descriptors computed directly from the input coherency matrix $\mathbf{T}_3$. The channel dimensions double at each stage:

$$C_1 = 16, \quad C_2 = 32, \quad C_3 = 64, \quad C_4 = 128$$

This progression balances feature richness with computational efficiency while capturing multi-scale polarimetric patterns.

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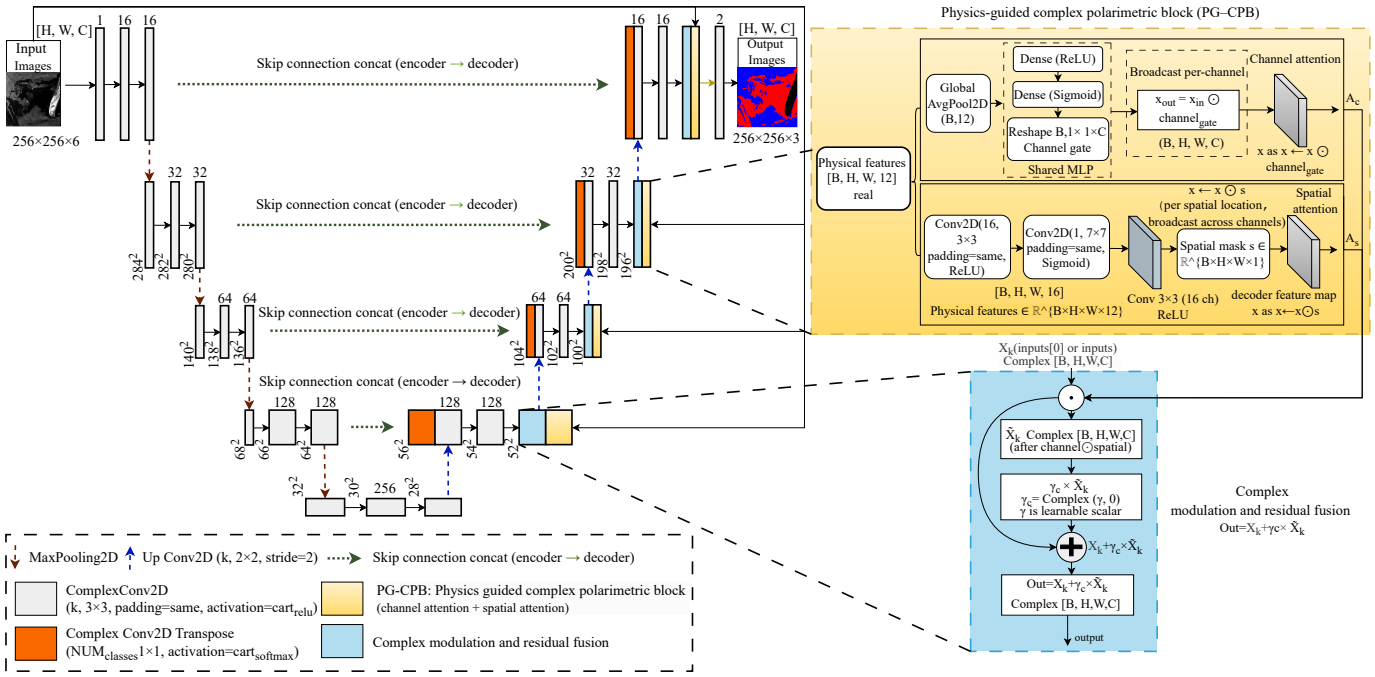


Fig. 3: Schematic of the proposed CPA-Net. The network employs a complex-valued U-Net backbone with skip connections, where the PG-CPB at each decoder stage injects polarimetric features to generate channel and spatial attention maps. The resulting attention responses modulate complex decoder features and are fused with a learnable complex residual path for polarimetric-aware feature refinement.

C. Bottleneck: High-Level Representation with Transformer Context

At the lowest resolution, the encoder output $\mathbf{P}_4$ is first mapped to a complex bottleneck feature by two complex convolutional layers:

$$\mathbf{B} = \phi(\mathbf{W}_{B,2} * \phi(\mathbf{W}_{B,1} * \mathbf{P}_4)), \qquad \mathbf{B} \in \mathbb{C}^{H_b \times W_b \times C_B} \tag{20}$$

where $C_B = 256$ and $\phi(\cdot)$ denotes the complex nonlinearity (e.g., cartesian ReLU).

Complex-to-Real Token Interface: To apply a real-valued Transformer while preserving phase-related information, a real-valued representation by concatenating the real part, the imaginary part, and the magnitude is built:

$$\mathbf{B}_{\text{real}} = \Big[\Re(\mathbf{B}),\ \Im(\mathbf{B}),\ |\mathbf{B}|\Big] \in \mathbb{R}^{H_b \times W_b \times 3C_B} \tag{21}$$

Then, $\mathbf{B}_{\text{real}}$ is embedded into a token sequence (using a 1×1 projection and flattening):

$$\mathbf{U} = \text{Flatten}(\mathbf{W}_e * \mathbf{B}_{\text{real}}) \in \mathbb{R}^{N \times D}, \qquad N = H_b W_b \tag{22}$$

D is the Transformer embedding dimension. Then add positional embeddings $\mathbf{E} \in \mathbb{R}^{N \times D}$:

$$\mathbf{U}_0 = \mathbf{U} + \mathbf{E} \tag{23}$$

Transformer Context Modeling: A Transformer block $\mathcal{T}(\cdot)$ (multi-head self-attention + MLP with residual connections) produces context-enhanced tokens:

$$\mathbf{U}_{\text{ctx}} = \mathcal{T}(\mathbf{U}_0) \in \mathbb{R}^{N \times D} \tag{24}$$

The tokens are reshaped back to a feature map and projected to a real-valued modulation tensor:

$$\mathbf{M}_{\text{real}} = \mathbf{W}_m * \text{Unflatten}(\mathbf{U}_{\text{ctx}}) \in \mathbb{R}^{H_b \times W_b \times C_B} \tag{25}$$

Complex Modulation and Residual Update: Finally, the Transformer output is used to modulate the complex bottleneck feature. Using a sigmoid gate,

$$\mathbf{G} = \sigma(\mathbf{M}_{\text{real}}) \in (0,1)^{H_b \times W_b \times C_B} \tag{26}$$

The sigmoid function $\sigma(\cdot)$ produces the gating tensor $\mathbf{G} \in (0,1)^{H_b \times W_b \times C_B}$ which, modulating the complex bottleneck feature through element-wise multiplication $\odot$, is applied identically to the real and imaginary components. The updated bottleneck is

$$\widetilde{\mathbf{B}} = \mathbf{B} \odot \mathbf{G}, \qquad \mathbf{B}' = \mathbf{B} + \lambda\, \widetilde{\mathbf{B}} \tag{27}$$

where the scalar λ is a learnable (or fixed) residual scaling factor. This yields a globally context-aware complex bottleneck $\mathbf{B}'$ that is passed to the decoder.

D. Decoder: Physics-Assisted Feature Refinement

The decoder reconstructs spatial resolution while integrating physics-assisted attention at each level. For each stage $k \in \{4, 3, 2, 1\}$:

$$\mathbf{X}_k = \phi_{\text{ReLU}}\left(\mathbf{W}_{k,2} * \left(\phi_{\text{ReLU}}(\mathbf{W}_{k,1} * [\mathcal{U}(\mathbf{U}_{k+1}) \| \mathbf{E}_k])\right)\right) \tag{28}$$

$$\mathbf{U}_k = \text{PG-CPB}(\mathbf{X}_k, \mathbf{X}_0) \tag{29}$$

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where $\mathcal{U}(\cdot)$ denotes complex transposed convolution with 2×2 kernels and stride 2, and $\|$ indicates channel-wise concatenation. Decoder features combine high-level semantic context with spatial detail from skip connections. Modulating them with physics-assisted attention focuses refinement on regions exhibiting characteristic oil-spill and look-alikes signatures.

E. Physics-Assisted Modulation

The PG-CPB module ingests two inputs: the complex feature map $\tilde{\mathbf{U}}_k$ and the original coherency tensor $\mathbf{X}_0$. From $\mathbf{X}_0$, it computes physical features f containing polarimetric ratios and complex coherences (as defined in Section III). Two attention mechanisms are derived:

$$\mathbf{A}_c = \sigma\left(\mathbf{W}_2\delta\left(\mathbf{W}_1\mathrm{GAP}(f)\right)\right) \tag{30}$$

$$\mathbf{A}_s = \sigma\left(\mathrm{Conv}_{7\times7}\left(\delta\left(\mathrm{Conv}_{3\times3}(f)\right)\right)\right) \tag{31}$$

where $\mathbf{A}_c \in [0,1]^{1\times1\times C}$ is the channel attention map, and $\mathbf{A}_s \in [0,1]^{H\times W\times1}$ is the spatial attention map.

These attention maps modulate the complex features:

$$\tilde{\mathbf{X}}_k = \mathbf{X}_k \odot \mathbf{A}_s \odot \mathbf{A}_c \tag{32}$$

followed by residual fusion:

$$\mathbf{U}_k = \mathbf{X}_k + \gamma\tilde{\mathbf{X}}_k \tag{33}$$

where $\gamma \in \mathbb{C}$ is a learnable complex scalar initialized to zero.

F. Output Projection

The final decoder output $\mathbf{U}_1$ is projected to class probabilities via 1×1 complex convolution:

$$\widehat{\mathbf{Y}} = \sigma_{\text{softmax}}\left(\mathbf{W}_O * \mathbf{U}_1 + \mathbf{b}_O\right) \tag{34}$$

where σ_{softmax} applies softmax to the magnitudes of complex logits, ensuring valid probability distributions.

G. Overall Formulation

The complete CPA-Net implements:

$$\widehat{\mathbf{Y}} = \Psi_{1\times1}\left(\mathcal{D}\left(\mathcal{E}(\mathbf{X}_0), \mathbf{X}_0\right)\right) \tag{35}$$

where $\mathcal{E}(\cdot)$ and $\mathcal{D}(\cdot)$ denote the encoder and decoder functions, and $\Psi_{1\times1}$ represents the final projection layer. By embedding PG-CPB within the decoder, CPA-Net maintains the representational power of complex-valued networks while enforcing physics-driven feature selection grounded in polarimetric scattering theory.

H. Feature Map Sizes and Channels

The proposed CPA-Net follows a symmetric encoder–decoder structure in which spatial resolution gradually decreases while the number of feature channels increases. The encoder begins with the complex-valued input $\mathbf{X}_0 \in \mathbb{C}^{p\times p\times6}$ and applies successive convolution–pooling stages, reducing the spatial size from $p\times p$ to $\frac{p}{16}\times\frac{p}{16}$ while expanding the channel depth from 16 to 256 as shown in Table I. The bottleneck at 256 channels forms the most semantically rich representation. The decoder symmetrically upsamples the feature maps back to full resolution through transposed convolutions and skip connections, progressively reducing channel dimensionality ($128 \rightarrow 16$) to refine spatial detail while preserving the high-level semantics extracted by the encoder. Finally, a 1×1 complex convolution produces the output of the segmentation $p\times p\times K$. This structured progression ensures efficient fusion of local polarimetric cues and global contextual dependencies while maintaining spatial fidelity crucial for delineating thin oil slick boundaries and small look-alike regions.

TABLE I: Network resolution and channel progression of the proposed CPA-Net. Here p denotes the input patch size and K denotes the number of output classes.

Stage	**Spatial Size**	**Channels**
Input $\mathbf{X}_0$	$p\times p$	6
Encoder Stage 1	$p\times p$	16
Pooling 1	$\frac{p}{2}\times\frac{p}{2}$	16
Encoder Stage 2	$\frac{p}{2}\times\frac{p}{2}$	32
Pooling 2	$\frac{p}{4}\times\frac{p}{4}$	32
Encoder Stage 3	$\frac{p}{4}\times\frac{p}{4}$	64
Pooling 3	$\frac{p}{8}\times\frac{p}{8}$	64
Encoder Stage 4	$\frac{p}{8}\times\frac{p}{8}$	128
Pooling 4	$\frac{p}{16}\times\frac{p}{16}$	128
Bottleneck $\mathbf{B}$	$\frac{p}{16}\times\frac{p}{16}$	256
Decoder Stage 4	$\frac{p}{8}\times\frac{p}{8}$	128
Decoder Stage 3	$\frac{p}{4}\times\frac{p}{4}$	64
Decoder Stage 2	$\frac{p}{2}\times\frac{p}{2}$	32
Decoder Stage 1	$p\times p$	16
Output Head	$p\times p$	K

V. Dataset and Experimental Setup

A. Quad-Polarimetric Oil-Spill Detection

To exploit the complete polarimetric scattering behavior of marine surfaces, this study utilizes the QPOSD dataset [41]. The dataset consists of UAVSAR multi-look complex (MLC) products which including complex-induced channels, are fundamental for capturing the necessary amplitude-phase relationships. The dataset was constructed to support the development and evaluation of physics-aware segmentation models for oil-spill detection and look-alike discrimination.

The dataset was acquired using the NASA Jet Propulsion Laboratory's UAVSAR platform, equipped with a fully polarimetric L-band SAR instrument operating since 2009. All MLC products were downloaded from the Alaska Satellite Facility (ASF) open-access portal and correspond to acquisitions from the UAVSAR gulf oil-spill campaign [56]. The imagery covers multiple time frames: *pre-spill* (2009–early 2010), *during-spill* (June 2010, Deepwater Horizon event), and *post-spill* (June 2011–July 2022). This temporal coverage ensures inclusion of both actual oil-contaminated and natural

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TABLE II: Specifications of the quad-polarization oil-spill detection dataset

Feature	Details
Dataset Name	Quad-Pol oil-spill detection (QPOSD)
Sensor Platform	Uninhabited aerial vehicle SAR (UAVSAR)
Radar Band	L-band (23.8 cm wavelength)
Product Type	Multi-look complex (MLC)
Spatial Resolution	6 m (azimuth) $\times$ 4 m (range)
Incidence Angle Range	27°–67°
Polarimetric Channels	HH, HV, VH, VV
Number of Image Samples	41 (from 15 scenes)
Classes	Sea Surface, oil-spill, Look-alike, Land
Label Format	Pixel-wise segmentation masks (.png)
Acquisition Period	June 2010 – June 2022
Geographical Region	Gulf of Mexico, USA

oceanic scenes for comprehensive generalization testing. The QPOSD dataset encompasses a wide range of environmental conditions, including variable sea states, wind speeds, and oil-spill morphologies. This diversity allows for effective model evaluation across complex scattering scenarios. Each image was manually annotated into four categories: sea surface, oil-spill, look-alike (low-wind or biogenic film areas), and land. The pixel-level labels employ the color scheme summarized in Table III.

TABLE III: Segmentation classes, RGB colour values, one-hot label indices, and pixel distributions in the QPOSD dataset. [36]

Class	RGB Value	One-Hot	Pixels	Percentage
Sea Surface	(0, 0, 255) Blue	0	61,118,506	76.44%
Oil-spill	(255, 0, 0) Red	1	12,478,428	15.61%
Look-alike	(0, 255, 0) Green	2	4,780,381	5.98%
Land	(0, 0, 0) Black	3	1,583,762	1.98%
Total	–	–	**79,961,077**	**100%**

Although the dataset does not include in-situ field measurements, all SAR scenes have been validated by NASA/JPL and cross-checked against previously published oil-spill studies in the Gulf of Mexico. This indirect validation ensures the authenticity and reliability of the labeled scenes while acknowledging the absence of ground-based sampling as a limitation. The dataset is publicly available at https://dx.doi.org/10.21227/rqyj-fj45, https://doi.org/10.5281/zenodo.19258036 [57]. The UAVSAR MLC data were analyzed through the own 3×3 polarimetric coherency matrix $\mathbf{T}_3$, which captures both power and correlation information among polarization channels. Representative scenes of the dataset are shown in Fig. 4

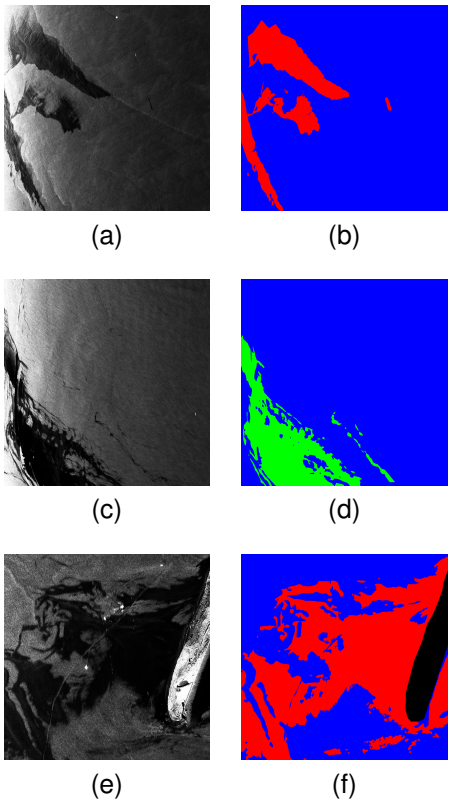


Fig. 4: Examples of UAVSAR L-band scenes (left column) and their corresponding manually annotated masks (right column). The color-code used for the segmentation classes is as follows: red (oil-spill), green (look-alike), blue (sea surface), and black (land). The middle scene illustrates a low-wind region (look-alike), while the top and bottom scenes contain verified oil-spills.

B. Experimental Setup

The QPOSD image samples were first divided into fixed-size patches of 256×256 pixels. After patch extraction, the resulting patches were randomly divided into training and testing sets using an 80:20 ratio, with a fixed random seed of 60 to ensure reproducibility. Therefore, the dataset partitioning in this study was performed at the patch level rather than at the original scene level. To ensure a fair comparison, all baseline methods and the proposed CPA-Net were evaluated using the same polarimetric input information and training protocol. No data augmentation was applied during training. All models were trained for a maximum of 100 epochs with a batch size of 8 using the Adam optimizer with an initial learning rate of 1×10^{-4}. Early stopping with a patience of 12 epochs was applied consistently to all models. No model-specific hyperparameter search was performed, and the remaining training hyperparameters were kept identical across all compared methods.

The CPA-Net was trained end-to-end using the Adam optimizer with an initial learning rate of $\eta = 1 \times 10^{-4}$. A batch size of 8 was employed for all experiments to balance convergence stability and GPU memory constraints. The optimizer parameters were set as $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, following the standard Adam formulation:

$$\theta_{t+1} = \theta_t - \eta, \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon} \tag{36}$$

where $\widehat{m}_t$ and $\widehat{v}_t$ are bias-corrected first- and second-moment

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estimates of the complex-valued gradients.

C. Loss Function

The training objective was the *complex average cross-entropy (CACE)* loss, which generalizes categorical cross-entropy to complex-valued outputs [58]. For a target distribution $\mathbf{y} \in \{0,1\}^K$ and a complex network output $\mathbf{z} \in \mathbb{C}^K$, the probability of class k is defined using magnitude normalization:

$$p_k = \frac{\exp(|z_k|)}{\sum_{j=1}^{K} \exp(|z_j|)} \tag{37}$$

The use of the magnitude $|z_k|$ instead of a full complex-valued softmax is deliberate. First, class probabilities must be non-negative and sum to one; the magnitude $|z_k|$ naturally provides a real-valued activation strength for each class. Second, applying the softmax operation to the magnitude preserves the phase information in the intermediate complex-valued representations, thereby maintaining their physical interpretability as indicators of scattering-mechanism. In contrast, a complex softmax based on full complex exponentials would entangle phase information with probability estimation and would compromise this interpretability, which is one of the key design objectives of CPA-Net. The CACE loss is given by:

$$\mathcal{L}_{\text{CACE}} = -\frac{1}{N} \sum_{n=1}^{N} \sum_{k=1}^{K} y_{n,k} \, \log(p_{n,k}) \tag{38}$$

which corresponds to the real component of the negative complex log-likelihood averaged over N pixels. This formulation ensures stable training by back-propagating gradients only through the magnitude of complex logits, preserving phase consistency in intermediate representations.

D. Evaluation Metrics

The quantitative performance of all models was evaluated using multiple complementary metrics to provide a balanced assessment of pixel-level segmentation accuracy. These include: complex-aware categorical accuracy, per-class intersection-over-Union (IoU), F1-score, precision, recall, and the Cohen's Kappa coefficient.

(1) Complex-Categorical Accuracy: the overall accuracy is given by

$$\text{Acc} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}(\hat{k}_n = k_n^*) \tag{39}$$

where $\mathbf{1}(\cdot)$ denotes the indicator function and N is the total number of pixels, $\hat{k}_n = \arg\max_k |z_{n,k}|$ denote the predicted class index for pixel n, where $z_{n,k}$ is the complex-valued network output for class k, and k_n^* be the corresponding ground-truth label.

(2) Per-Class Intersection-over-Union: for each class c, the IoU is defined as

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{40}$$

where TP_c, FP_c, and FN_c represent the numbers of true-positive, false-positive, and false-negative pixels of class c, respectively. The mean IoU (mIoU) over all K classes is computed as

$$\text{mIoU} = \frac{1}{K} \sum_{c=1}^{K} \text{IoU}_c \tag{41}$$

(3) Precision and Recall: The pixel-wise precision and recall for each class c are defined as

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{42}$$

The first measures the correctness of the predicted positive pixels, while the second reflects the proportion of actual positives correctly identified.

(4) F1-Score: The F1-score provides a harmonic mean between precision and recall, yielding a balanced measure of classification accuracy for each class:

$$\text{F1}_c = \frac{2 \times \text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{43}$$

The overall F1-score is reported as the arithmetic mean over all classes.

(5) Cohen's Kappa Coefficient: To quantify the agreement between the predicted and ground-truth classifications beyond that expected from the class distribution by chance, the Cohen's Kappa coefficient (κ) is computed as

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{44}$$

where p_o represents the observed agreement between the predicted and ground-truth classifications, corresponding to the overall accuracy, while p_e represents the agreement expected by chance based on the marginal class distributions of the confusion matrix. Thus, κ measures the degree of agreement beyond that attributable to the class distribution alone. A κ value of 1 indicates perfect agreement, while values below 0 imply agreement worse than random chance.

Together, these metrics comprehensively evaluate both the pixel-level classification consistency and the segmentation quality across all oil-spill and look-alike categories in the QPOSD dataset. All complex operations were implemented using the CVNN framework with `ComplexHeNormal` initialization and `cart_ReLU` activation functions. This configuration ensured numerically stable convergence while preserving the amplitude–phase relationships crucial for polarimetric scattering analysis.

VI. Experimental Results

A. Quantitative Evaluation

For comparative evaluation, several representative baseline networks covering different architectural paradigms were considered, including real-valued, complex-valued, attention-based, and transformer-based models. Specifically, the baselines include CDA-Net, MSHFF-Net, OSDTAU-Net, CBAM-U-Net, CWCM-Net (with and without multi-pol feature fusion) and U-ViT. These methods represent recent state-of-the-art approaches for SAR-based oil spill segmentation, incorporating multi-scale feature fusion, attention mechanisms, and transformer modules.

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TABLE IV: Quantitative comparison of baseline networks on the QPOSD dataset

Model	**Acc**	**mIoU**	**F1**	**Precision**	**Recall**	κ	**Params (M)**	**GFLOPs**(256^2)
CDA-Net [59]	0.9724	0.8867	0.9337	0.9497	0.9337	0.9282	10.9	34.77
MSHFF-Net [60]	0.9767	0.8895	0.9410	0.9495	0.9332	0.9385	6.66	17.34
OSDTAU-Net [61]	0.9880	0.9453	0.9722	0.9730	0.9713	0.9686	19.2	11.30
CBAM-U-Net [62]	0.9621	0.8642	0.9123	0.9314	0.8983	0.9043	2.05	6.24
CWCM-Net [41]	0.9586	0.8608	0.9004	0.9174	0.9042	0.9031	0.51	0.21
CWCM-Net (Feature-Fusion) [41]	0.9727	0.8875	0.9269	0.9402	0.9124	0.9204	4.39	13.64
U-ViT [63]	0.9758	0.8964	0.9452	0.9646	0.9289	0.9355	1.85	5.74
CPA-Net	**0.9900**	**0.9556**	**0.9755**	**0.9819**	**0.9734**	**0.9739**	4.56	25.58

The proposed CPA-Net adopts a fully complex-valued U-Net backbone augmented with physics-assisted polarimetric attention, enabling joint exploitation of amplitude, phase, and polarimetric coherence information. All models were trained and evaluated under identical experimental settings to ensure a fair comparison. As shown in Table IV, CPA-Net achieves the best overall performance across all reported metrics. It attains the highest overall accuracy of 0.9900, mIoU of 0.9556, F1-score of 0.9755, and Cohen's κ coefficient of 0.9739, indicating strong segmentation accuracy and excellent agreement with ground truth labels. The table also provides the parameter count in millions (Params(M)) and Giga Floating-Point Operations Per Second (GFLOPs). A closer examination of the results reveals several important observations:

- Overall Performance Gains: CPA-Net consistently outperforms all baseline models, particularly in mIoU, which provides a stricter and more class-balanced evaluation of segmentation quality. Compared with the strongest competing method, OSDTAU-Net, CPA-Net improves mIoU from 0.9453 to 0.9556, demonstrating the benefit of explicitly integrating polarimetric physics into the learning process.
- Impact of Physics-Assisted Polarimetric Attention: The performance differences among the evaluated methods indicate that architectural complexity alone does not guarantee improved segmentation accuracy. Several strong baselines, including transformer-based U-ViT and multi-scale fusion models CDA-Net and MSHFF-Net, already achieve competitive results, particularly in overall accuracy. However, their performance saturates in terms of mIoU and κ. By contrast, CPA-Net consistently improves mIoU and class-balanced metrics, highlighting the advantage of explicitly incorporating polarimetric scattering characteristics into the attention mechanism. The physics-assisted polarimetric attention selectively emphasizes regions exhibiting physically meaningful damping, depolarization, and coherence behavior, which are not directly enforced in generic attention or transformer modules. This targeted feature modulation leads to more reliable discrimination between oil spills, look-alikes, and clean sea surfaces, resulting in measurable gains across all reported metrics.
- Balance Between Accuracy and Model Complexity: Although CPA-Net is not the smallest model, with 4.56M parameters it remains competitive in terms of complexity. It achieves superior performance compared to larger models such as OSDTAU-Net with 19.2M parameters and CDA-Net with 10.9M parameters, while maintaining a moderate parameter count. This indicates that the observed performance improvements are not merely due to increased model capacity, but rather to more effective feature representation and attention design.

The quantitative evaluation demonstrates that CPA-Net provides a meaningful solution for quad-polarimetric oil spill segmentation. The results confirm that incorporating physics-assisted polarimetric attention leads to consistent improvements in segmentation accuracy and reliability, while maintaining a favorable balance between performance and computational efficiency.

B. Qualitative Comparison of Segmentation Results

Fig. 5 presents a qualitative comparison of segmentation results produced by different baseline models and the proposed CPA-Net on representative scenes from the QPOSD dataset. The selected examples cover diverse scenarios, including coastal regions with land–sea boundaries, open-water oil spill events, and challenging look-alike phenomena. The visual comparison enables an intuitive assessment of how well each model delineates the four target classes: sea surface (blue), oil spill (red), look-alike (green) and land (black).

In the first row, which consists of a coastal area, look-alike phenomena, and sea surface, CPA-Net and CDA-Net produce segmentation results that are visually closest to the ground truth, with clearer class separation and more consistent spatial patterns. In contrast, OSTDAU-Net, CWCM-Net (FF), and CBAM-Net show limitations in accurately segmenting look-alike regions, particularly near coastal transitions. MSHFF-Net exhibits pronounced difficulties across all three classes, with unstable boundaries and noticeable confusion at class edges. U-ViT and the final CWCM-Net under-segment look-alike areas and additionally introduce false alarms by incorrectly predicting oil spill regions that are not present in the scene.

In the second row, which contains only oil spill and sea surface, most models demonstrate improved performance, reflecting the reduced class complexity. CPA-Net, CWCM-Net (FF), and OSTDAU-Net provide relatively consistent oil spill delineation. However, MSHFF-Net, U-ViT, and CBAM-Net struggle to accurately capture the oil spill extent, exhibiting

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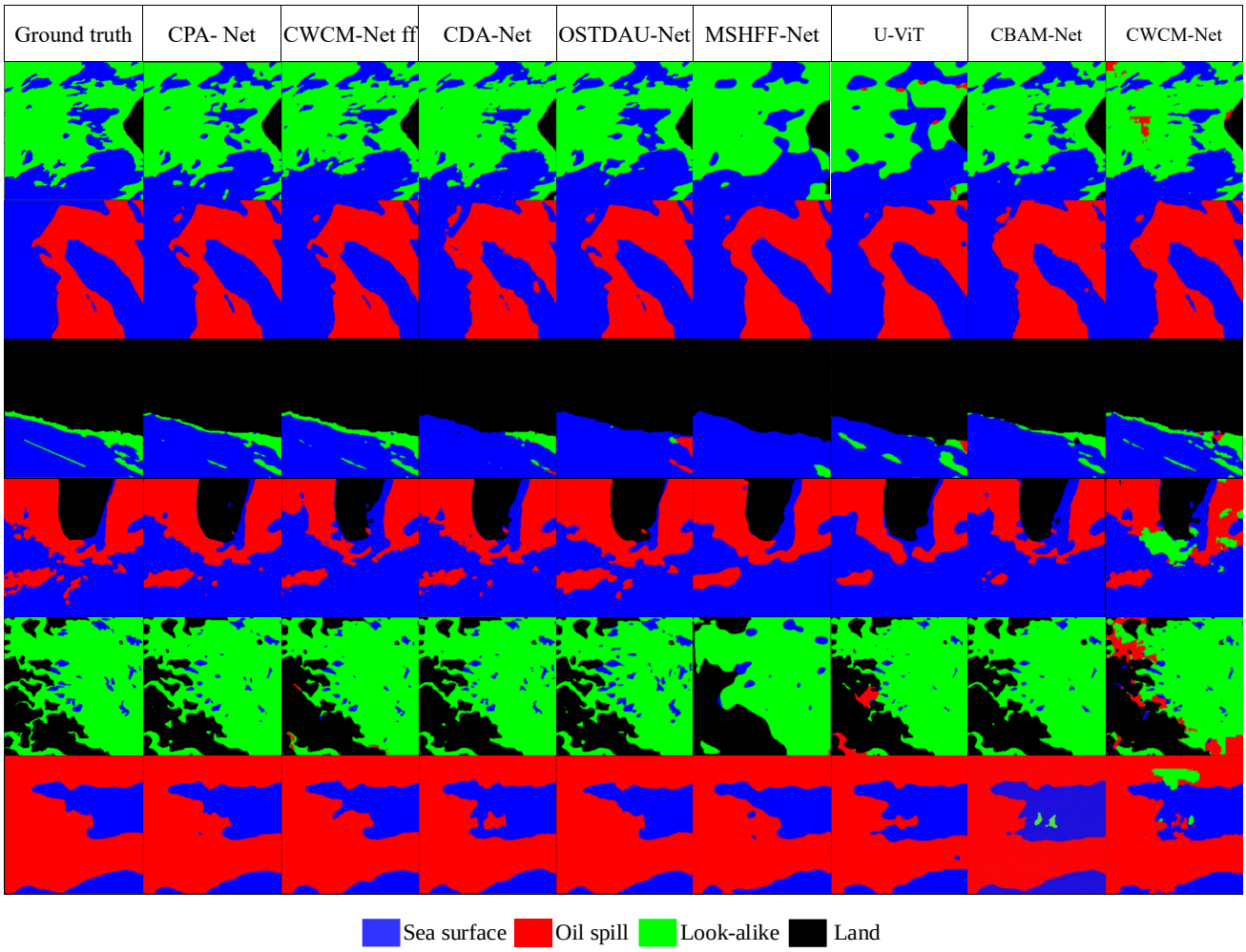


Fig. 5: Qualitative comparison of segmentation results on representative QPOSD scenes. Different rows correspond to varying class compositions and scene complexities, illustrating the relative robustness of CPA-Net against baseline models.

boundary irregularities and partial misclassification of surrounding sea surface regions.

The third row again represents a coastal scene composed of land, look-alike phenomena, and sea surface. In this case, CPA-Net and CWCM-Net (FF) maintain close agreement with the ground truth, preserving both class extent and spatial continuity. The remaining models show substantial degradation in performance. Notably, OSTDAU-Net, U-ViT, and the final CWCM-Net introduce false oil spill predictions, indicating poor discrimination between look-alike phenomena and oil-free sea surface areas.

In the fourth row, which includes land, oil spill, and sea surface, CPA-Net yields the most accurate segmentation, with coherent oil spill regions and well-defined land–water boundaries. Other models tend to under-segment the oil spill class and exhibit imprecise land boundaries. Additionally, CWCM-Net misclassifies portions of the oil spill as look-alike phenomena, further reducing segmentation consistency.

The fifth row highlights another coastal scene dominated by look-alike phenomena and sea surface. CPA-Net, CDA-Net, OSTDAU-Net, and CBAM-Net achieve visually satisfactory segmentation results, maintaining stable class separation. In contrast, MSHFF-Net continues to underperform, particularly at class boundaries. CWCM-Net (FF) introduces localized misclassification of oil spill in small regions. U-ViT and the final CWCM-Net perform the poorest in this scenario, producing extensive misclassification across multiple classes.

In the sixth row, which consists of oil spill and sea surface only, CPA-Net, CWCM-Net (FF), and OSTDAU-Net demonstrate comparatively better performance, yielding more coherent oil spill predictions. Conversely, CBAM-Net and the final CWCM-Net perform poorly, misclassifying several regions as look-alike phenomena and introducing notable segmentation noise.

Overall, the qualitative comparisons demonstrate that CPA-Net consistently provides more stable and physically consistent segmentation results across diverse scene compositions, including coastal environments, look-alike–dominated regions, and open-sea oil spill scenarios. While several baseline models perform reasonably under reduced class complexity, their performance degrades in complex coastal scenes, often exhibiting boundary instability, class confusion, or false alarms. In con-

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TABLE V: Class-wise precision (P), recall (R), and F1-score (F1) comparison of different segmentation methods on the QPOSD dataset.

Method	Sea Surface			Oil Spill			Look-alike			Land		
	P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
CDA-Net	0.9882	0.9838	0.9860	0.9084	0.9737	0.9399	0.9523	0.8214	0.8820	0.9497	0.9557	0.9527
MSHFFNet	0.9834	0.9911	0.9872	0.9733	0.9628	0.9680	0.9016	0.8313	0.8650	0.9396	0.9476	0.9436
OSDTAU-Net	0.9924	0.9947	0.9936	0.9870	0.9751	0.9810	0.9348	0.9358	0.9353	0.9777	0.9798	0.9788
CBAM	0.9827	0.9853	0.9840	0.8907	0.9484	0.9186	0.9052	0.7459	0.8179	0.9438	0.9141	0.9287
CWCM-Net	0.9874	0.9842	0.9858	0.8923	0.9509	0.9207	0.8629	0.7249	0.7879	0.9243	0.9577	0.9407
U-ViT	0.9786	0.9932	0.9859	0.9716	0.9548	0.9631	0.9499	0.8030	0.8703	0.9581	0.9647	0.9614
CPA-Net	0.9923	0.9937	0.9930	0.9854	0.9850	0.9852	0.9580	0.9367	0.9472	0.9708	0.9821	0.9764

trast, CPA-Net shows improved robustness to scene variability and class ambiguity, leading to more reliable delineation of oil spill, look-alike, sea surface, and land regions, which is consistent with the observed quantitative trends.

C. Per-Class Performance Analysis

Table V reports the class-wise precision, recall, and F1-score of the evaluated methods, providing a more detailed assessment of their performance across both dominant and minority classes. CPA-Net achieves particularly strong performance for the two marine target classes, with precision, recall, and F1-score of 0.9854, 0.9850, and 0.9852 for oil spills, respectively, and 0.9580, 0.9367, and 0.9472 for look-alikes. Although the look-alike class remains more challenging than the oil-spill class, CPA-Net achieves the highest precision, recall, and F1-score for look-alikes among the compared methods. In particular, compared with CWCM-Net, the look-alike F1-score increases from 0.7879 to 0.9472, indicating substantially improved discrimination of oil-like phenomena.

Fig. 6 further presents the per-class IoU comparison. Together with the precision, recall, and F1-score results in Table V, the class-wise evaluation reveals distinct performance characteristics across the four semantic categories.

- **Sea Surface:** All evaluated models achieve consistently high performance for the sea-surface class, reflecting its relatively homogeneous scattering characteristics. CPA-Net achieves an IoU of 0.9861 together with precision, recall, and F1-score of 0.9923, 0.9937, and 0.9930, respectively. The relatively small differences among the strongest methods indicate that sea-surface segmentation is less challenging than discrimination between oil spills and look-alikes.
- **Oil Spill:** Greater performance variation is observed for the oil-spill class. CWCM-Net and CBAM achieve IoUs of 0.8530 and 0.8495, respectively, whereas CPA-Net increases the IoU to 0.9708. CPA-Net also achieves a precision of 0.9854, recall of 0.9850, and F1-score of 0.9852. These results indicate that incorporating physically meaningful polarimetric descriptors into the feature-learning process improves the discrimination of oil-spill regions.
- **Look-Alike:** Look-alikes constitute the most challenging class, with substantially lower performance for several baseline methods. For example, CWCM-Net obtains an IoU of 0.6500 and an F1-score of 0.7879, while CBAM achieves an IoU of 0.6918 and an F1-score of 0.8179. In contrast, CPA-Net achieves an IoU of 0.8997, precision of 0.9580, recall of 0.9367, and F1-score of 0.9472. Moreover, relative to the baseline U-Net, the look-alike IoU increases from 0.6627 to 0.8997, corresponding to an absolute improvement of 23.70 percentage points, which is larger than the improvement observed for the oil-spill class (0.8425 to 0.9708, +12.83 percentage points). This result indicates that the proposed physics-guided framework provides its largest class-level benefit for the difficult look-alike discrimination problem.
- **Land:** Land regions are also reliably identified due to their comparatively distinct scattering characteristics. CPA-Net achieves an IoU of 0.9539, with precision, recall, and F1-score of 0.9708, 0.9821, and 0.9764, respectively. The strong land performance indicates that the proposed framework maintains effective discrimination across both marine and non-marine regions.

a) Overall Assessment: Overall, CPA-Net demonstrates a balanced performance across all four classes, with its most pronounced improvement observed for the challenging look-alike category. The simultaneous improvements in look-alike discrimination and oil-spill segmentation, while maintaining strong performance for sea-surface and land regions, demonstrate the effectiveness of incorporating physically meaningful polarimetric information into the segmentation framework.

VII. Feature Analysis and Ranking

A. Feature Importance: Oil vs. Look-alikes

To evaluate feature effectiveness in distinguishing oil-spill from look-alikes, a tailored analysis is conducted. Table VI presents the performance ranking of polarimetric features for oil versus look-alike discrimination. The magnitude of the complex correlation coefficient $|\gamma_{12}|$ once again ranks first, achieving the highest F1 score of 0.8604 and IoU of 0.7550. This consistent top performance across all discrimination tasks highlights its exceptional robustness even in the presence of look-alikes.

Notably, the CPR and the $R_{22/11}$ achieve very strong second and third positions with F1 scores of 0.8571 and 0.8558, respectively. This is particularly significant and agrees with

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Fig. 7 that shows the ability of these features in distinguishing look-alikes with respect to the surrounding area. Their high ranking here indicates that they are especially useful for suppressing false alarms caused by look-alikes. The strong performance of $|\gamma_{13}|$ and $|\gamma_{23}|$ (ranks 4 and 5) further reinforces the importance of inter-channel correlation information.

Phase-related features $\cos\angle\gamma_{12}$ and $\sin\angle\gamma_{12}$ also demonstrate competitive performance, occupying the 6th and 7th positions. This suggests that phase information provides complementary discriminative power that is particularly valuable when separating oil from look-alikes. In contrast, $R_{33/11}$ shows good but slightly lower performance compared to the top features.

It is worth noting that several phase features $\sin\angle\gamma_{23}$, $\cos\angle\gamma_{23}$, etc.) achieve very high recall (above 0.993) but lower precision, indicating a tendency to over-detect oil in this challenging scenario.

The results in Table VI reveal that while $|\gamma_{12}|$ remains the most reliable single feature, a combination of correlation magnitude, intensity ratios (especially $R_{22/11}$), and CPR is highly effective for the critical task of look-alike suppression as shown in Fig. 7. These findings provide important guidance for designing feature-level fusion strategies in the proposed deep learning models and validate the inclusion of physics-assisted polarimetric observables in CPA-Net and related architectures.

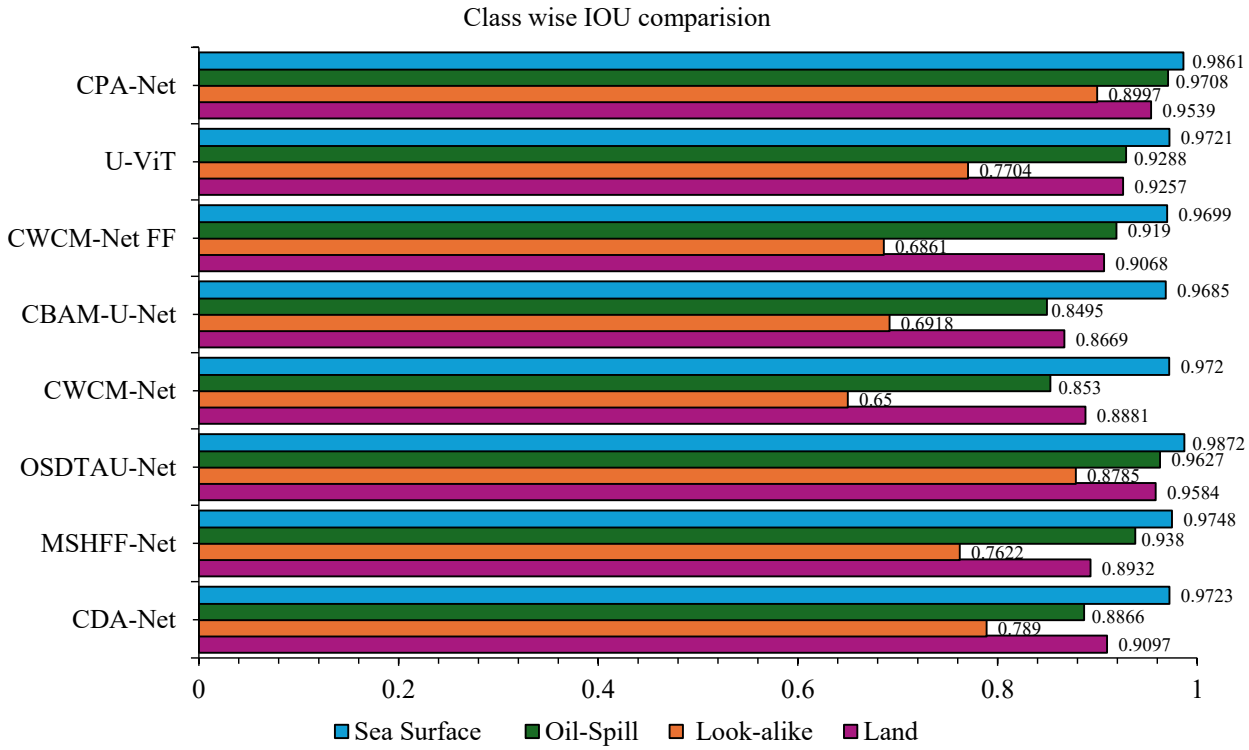


Fig. 6: Per-class IoU comparison of the evaluated segmentation methods on the QPOSD dataset.

TABLE VI: Top-ranked features for distinguishing oil spills from look-alikes based on classification performance

Rank	Feature Name	F1 Score	IoU	Accuracy	Precision	Recall
1	$\|\gamma_{12}\|$	0.8604	0.7550	0.7676	0.7666	0.9803
2	CPR	0.8571	0.7500	0.7639	0.7682	0.9693
3	$R_{22/11}$	0.8558	0.7480	0.7637	0.7721	0.9599
4	$\|\gamma_{13}\|$	0.8545	0.7460	0.7590	0.7642	0.9691
5	$\|\gamma_{23}\|$	0.8543	0.7457	0.7600	0.7675	0.9633
6	$\cos\angle\gamma_{12}$	0.8510	0.7407	0.7484	0.7499	0.9837
7	$\sin\angle\gamma_{12}$	0.8506	0.7401	0.7563	0.7703	0.9497
8	$R_{33/11}$	0.8502	0.7395	0.7589	0.7783	0.9369
9	$\sin\angle\gamma_{23}$	0.8439	0.7299	0.7314	0.7333	0.9938
10	$\cos\angle\gamma_{23}$	0.8436	0.7296	0.7310	0.7331	0.9935
11	$\cos\angle\gamma_{13}$	0.8436	0.7295	0.7309	0.7330	0.9934
12	$\sin\angle\gamma_{13}$	0.8434	0.7292	0.7305	0.7328	0.9932

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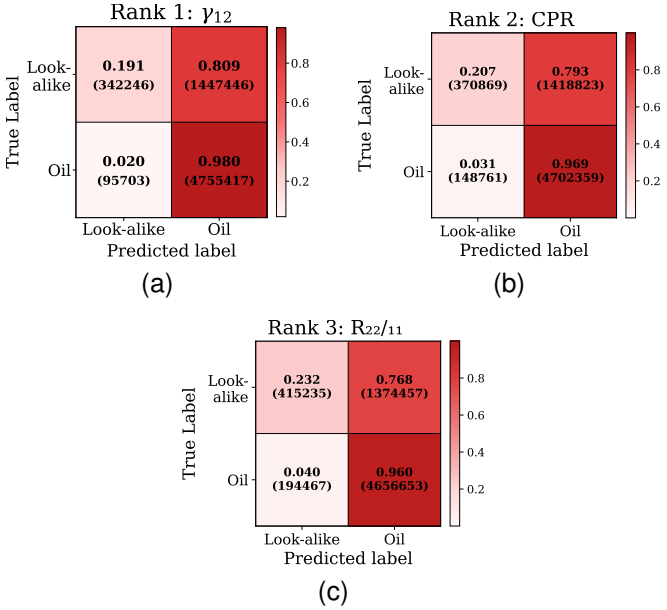


Fig. 7: Normalized confusion matrices summarizing oil and look-alike classification performance across top three physical descriptors that influence the training process. The matrices highlight correct predictions and cross-class confusion patterns.

B. Feature Importance: oil-water discrimination

A critical step in understanding the effectiveness of polarimetric features is evaluating their ability to discriminate oil spills specifically from clean sea surface. This scenario represents the most fundamental binary classification task in marine oil spill detection, where look-alikes are not present.

Table VII shows the ranking of polarimetric features for oil versus sea surface discrimination. The magnitude of the complex correlation coefficient $|\gamma_{12}|$ performs best, achieving an F1 score of 0.7931, IoU of 0.6571, and accuracy of 0.9333. This outstanding performance underscores the high sensitivity of $|\gamma_{12}|$ to the damping effect of oil on the sea surface, which alters the correlation between co-polarized channels. The next two positions are occupied by $|\gamma_{23}|$ and $|\gamma_{13}|$, further confirming that features based on the magnitude of complex correlation coefficients are highly effective for distinguishing oil from clean sea. These top-three features demonstrate substantially superior performance compared to the remaining features, suggesting that correlation information captures essential scattering mechanism differences induced by mineral oil films. Phase-related features $\sin\angle\gamma_{ij}$ and $\cos\angle\gamma_{ij}$ show moderate discriminative power, with $\sin\angle\gamma_{12}$ ranking fourth. While they give useful information, their performance is notably lower than the magnitude-based counterparts. This indicates that amplitude correlation carries more discriminative information than phase in the relatively homogeneous sea surface background. Interestingly, the CPR and the intensity ratio $R_{22/11}$ exhibit the lowest F1 scores despite achieving extremely high recall values (0.9868). Their very low precision (0.1680) reveals a strong tendency to generate false positives, limiting their standalone utility. The feature $R_{33/11}$ performs better than these two but still lags significantly behind the correlation-based features.

The results in Table VII clearly demonstrate the superiority of complex correlation magnitude features $|\gamma_{12}|$, $|\gamma_{23}|$, and $|\gamma_{13}|$) for oil-sea surface discrimination as shown in Fig. 8. These findings are consistent with established polarimetric SAR theory regarding the impact of oil slicks on surface scattering and provide strong justification for incorporating these features as key inputs in the proposed physics-assisted deep learning architectures.

C. Feature Importance: oil-lookalikes discrimination

To comprehensively assess the discriminative capability of polarimetric features across varying background complexities, an additional experiment was performed by merging sea surface and look-alike pixels into a single “non-oil” class. This binary classification task better reflects real-world operational scenarios where the primary goal is to separate oil spills from all other ocean surface phenomena.

TABLE VII: Top-ranked features for distinguishing oil spills from sea surface based on classification performance

Rank	Feature Name	F1 Score	IoU	Accuracy	Precision	Recall
1	$\|\gamma_{12}\|$	0.7931	0.6571	0.9333	0.8307	0.7588
2	$\|\gamma_{23}\|$	0.5543	0.3834	0.8192	0.4741	0.6672
3	$\|\gamma_{13}\|$	0.4994	0.3328	0.7968	0.4269	0.6015
4	$\sin\angle\gamma_{12}$	0.4541	0.2937	0.7829	0.3940	0.5357
5	$\cos\angle\gamma_{12}$	0.4424	0.2840	0.7727	0.3771	0.5350
6	$\cos\angle\gamma_{13}$	0.3808	0.2352	0.6135	0.2608	0.7053
7	$\cos\angle\gamma_{23}$	0.3652	0.2234	0.5752	0.2441	0.7253
8	$\sin\angle\gamma_{23}$	0.3639	0.2224	0.5454	0.2381	0.7716
9	$\sin\angle\gamma_{13}$	0.3386	0.2038	0.4773	0.2152	0.7938
10	$R_{33/11}$	0.3341	0.2005	0.7514	0.3045	0.3700
11	CPR	0.2871	0.1676	0.1741	0.1680	0.9868
12	$R_{22/11}$	0.2871	0.1676	0.1741	0.1680	0.9868

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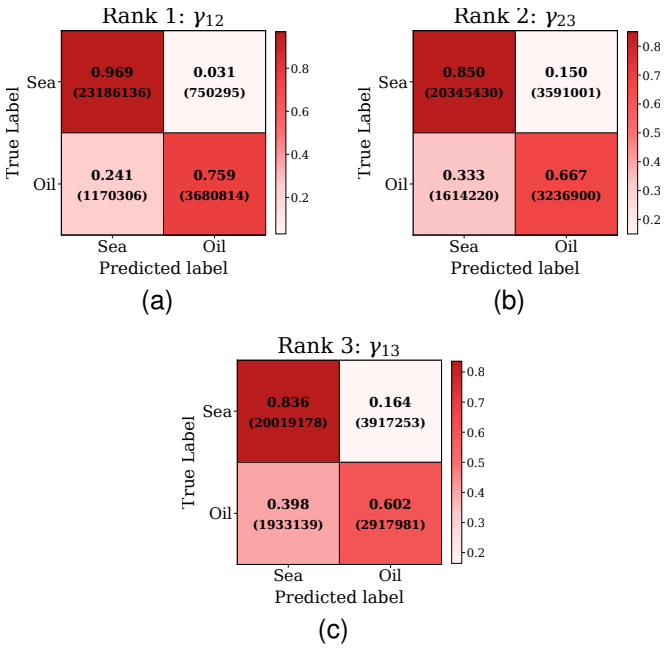


Fig. 8: Normalized confusion matrices summarizing oil and sea surface classification performance across top three physical descriptors that influence the training. The matrices highlight correct predictions and cross-class confusion patterns.

Table VIII presents the ranked performance of the extracted polarimetric features for oil versus non-oil discrimination. The results clearly indicate that the magnitude of the complex correlation coefficient between channels 1 and 2, $|\gamma_{12}|$, is the most powerful single feature. It achieves the highest F1 score (0.6867), Intersection over Union (IoU) of 0.5228, and overall accuracy of 0.8881. This consistent top ranking across all three discrimination tasks (oil-vs-sea, oil-vs-look-alike, and oil-vs-non-oil) demonstrates its robustness and importance in polarimetric SAR-based oil spill detection.

The second and third positions are occupied by $|\gamma_{23}|$ and $|\gamma_{13}|$, respectively. These findings reinforce the critical role of co-polarized and cross-polarized correlation information in capturing the scattering differences between oil-covered and oil-free surfaces. Features derived from the phase components ($\sin\angle\gamma_{ij}$ and $\cos\angle\gamma_{ij}$) show moderate performance, with $\sin\angle\gamma_{12}$ ranking fourth. This suggests that while phase information contributes valuable discriminative power, the magnitude of the correlation coefficients remains more informative for this particular application.

It is also observed that the ratio-based features ($R_{22/11}$ and $R_{33/11}$) and the CPR exhibit relatively lower performance in this setting. Notably, CPR and $R_{22/11}$ achieve very high recall (0.9874) but suffer from extremely low precision (0.1582), resulting in the lowest F1 scores. This behavior indicates that these features tend to over-predict oil, leading to a high number of false positives when distinguishing oil from the diverse non-oil backgrounds.

Overall, the feature ranking in the oil-versus-non-oil scenario maintains strong consistency with the previous analyses. The dominance of the $|\gamma_{ij}|$ features across all experiments highlights their potential as reliable inputs for subsequent physics-assisted deep neural network model as shown in Fig. 9. These findings provide valuable guidance for feature selection and model design, emphasizing the need to prioritize correlation magnitude features while carefully incorporating phase information to enhance robustness against look-alikes.

VIII. Ablation Study

To systematically assess the effectiveness of the proposed PG-CPB, we conducted a comprehensive ablation study on the QPOSD dataset. The experiments examine the individual contributions of phase, scattering power ratios, and coherence descriptors; isolate the effect of the physics-assisted attention mechanism; and evaluate their combined contribution to segmentation performance. All variants retain the same original $\mathbf{T}_3$ input representation and backbone architecture. Unless otherwise specified, the training strategy, optimizer, and hyperparameter settings are also kept consistent, with only the specified descriptor or attention components varied. This con-

TABLE VIII: Performance ranking of polarimetric features for oil spill versus non-oil discrimination

Rank	Feature	F1 Score	IoU	Accuracy	Precision	Recall
1	$\|\gamma_{12}\|$	0.6867	0.5228	0.8881	0.6178	0.7728
2	$\|\gamma_{23}\|$	0.4945	0.3285	0.7763	0.3854	0.6898
3	$\|\gamma_{13}\|$	0.4463	0.2873	0.7517	0.3453	0.6308
4	$\sin\angle\gamma_{12}$	0.4363	0.2790	0.7841	0.3724	0.5265
5	$\cos\angle\gamma_{12}$	0.4166	0.2631	0.7593	0.3385	0.5417
6	$\cos\angle\gamma_{13}$	0.3665	0.2244	0.6188	0.2489	0.6950
7	$\cos\angle\gamma_{23}$	0.3497	0.2119	0.5765	0.2311	0.7177
8	$\sin\angle\gamma_{23}$	0.3482	0.2108	0.5458	0.2254	0.7648
9	$\sin\angle\gamma_{13}$	0.3243	0.1935	0.4808	0.2043	0.7853
10	$R_{33/11}$	0.3064	0.1809	0.7112	0.2475	0.4022
11	CPR	0.2728	0.1579	0.1647	0.1582	0.9874
12	$R_{22/11}$	0.2728	0.1579	0.1647	0.1582	0.9874

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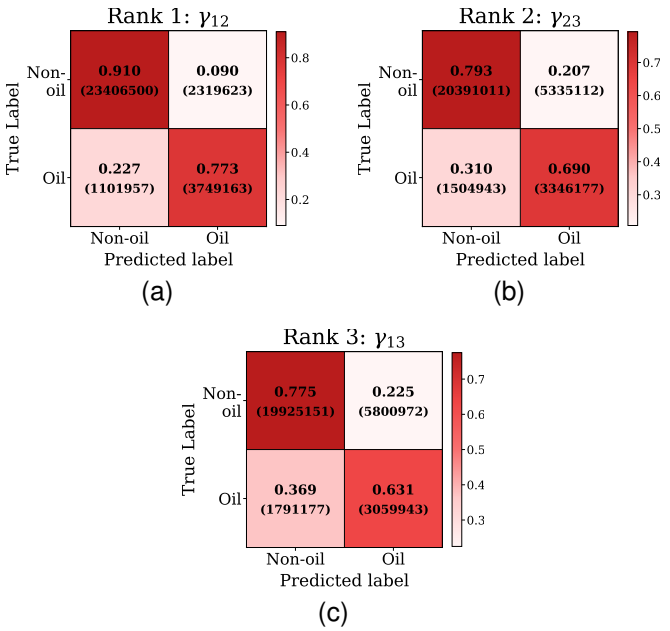


Fig. 9: Normalized confusion matrices summarizing oil and non-oil classification performance across top three physical descriptors that influence the training process. The matrices highlight correct predictions and cross-class confusion patterns.

trolled setup enables the observed performance differences to be primarily attributed to the components under investigation.

1) Contribution of Different Polarimetric Scattering Descriptors: The first four configurations in Table IX, including the baseline U-Net and the three single-descriptor variants, evaluate the individual contributions of phase, scattering power ratios, and coherence features. The baseline U-Net achieves an mIoU of 0.8497, an overall F1-score of 0.9100, and a Cohen's κ coefficient of 0.9034. Introducing phase descriptors improves the mIoU from 0.8497 to 0.8700 and the F1-score from 0.9100 to 0.9201. In particular, the IoU of the look-alike class increases from 0.6627 to 0.7258, representing an improvement of 6.31 percentage points. This result indicates that phase information provides useful complementary information about the relative phase relationships between polarimetric scattering channels, which can assist in distinguishing look-alike regions from other surface types. Nevertheless, the relatively limited improvement indicates that phase information alone is insufficient to fully characterize the scattering differences among oil spills, look-alikes, and sea surfaces. The use of scattering power-ratio descriptors produces a further improvement, with the mIoU increasing to 0.8811 and the F1-score to 0.9411. Compared with the baseline, the oil-spill IoU increases from 0.8425 to 0.8876, while the look-alike IoU increases from 0.6627 to 0.7704. These improvements suggest that power-ratio features provide discriminative information related to the relative distribution of polarimetric scattering power and can therefore complement the intensity information learned directly from the network input. Among the three individual descriptor groups, coherence features provide the strongest performance improvement. The PG-CPB (coherence) variant achieves an mIoU of 0.9006, an F1-score of 0.9511, and a Cohen's κ coefficient of 0.9350. The oil-spill IoU increases to 0.9095, while the look-alike IoU reaches 0.7900. Compared with phase and power-ratio descriptors alone, these results indicate that coherence provides particularly effective information about the statistical relationship between polarimetric scattering channels, thereby improving the discrimination of spectrally and structurally similar marine targets.

The results from the three individual descriptor variants demonstrate that phase, scattering power ratios, and coherence contribute complementary physical information. However, none of the individual descriptor groups achieves the performance obtained by jointly incorporating all physical descriptors.

When all physical descriptors are incorporated into the PG-CPB without the attention mechanism, the mIoU increases substantially to 0.9469, accompanied by an F1-score of 0.9728 and a Cohen's κ coefficient of 0.9696. Compared with the baseline U-Net, this corresponds to an mIoU improvement of 9.72 percentage points. The oil-spill IoU increases from 0.8425 to 0.9582, representing an improvement of 11.57 percentage points, while the look-alike IoU increases from 0.6627 to 0.8747, corresponding to an improvement of 21.20 percentage points. These results demonstrate that jointly exploiting phase, power-ratio, and coherence information provides substantially more discriminative representations than using any individual descriptor group alone. The particularly large improvement in the look-alike class suggests that the complementary physical descriptors are effective in capturing scattering characteristics that are difficult to distinguish using the learned image features alone.

2) Effectiveness of the Physics-Assisted Attention Mechanism: To evaluate the contribution of the proposed attention mechanism independently of the descriptor selection, the configuration using all physical descriptors without attention is compared with the corresponding model incorporating the proposed physics-assisted channel and spatial attention mechanisms. As shown in Table IX, introducing the attention mechanism further improves the mIoU from 0.9469 to 0.9556, while the F1-score increases from 0.9728 to 0.9755 and the Cohen's κ coefficient increases from 0.9696 to 0.9739.

The most notable improvements occur for the oil-spill and look-alike classes. The oil-spill IoU increases from 0.9582 to 0.9708, an improvement of 1.26 percentage points, whereas the look-alike IoU increases from 0.8747 to 0.8997, an improvement of 2.50 percentage points. In contrast, the IoUs of the sea-surface and land classes change slightly from 0.9889 to 0.9861 and from 0.9706 to 0.9539, respectively. Despite these small reductions for the two relatively well-separated classes, the improvements in the more challenging oil-spill and look-alike categories result in an overall mIoU gain of 0.87 percentage points.

These results indicate that the proposed attention mechanism complements the physical descriptors by adaptively emphasizing informative feature responses and suppressing less discriminative responses during feature aggregation. In particular, the larger improvement observed for the look-alike

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TABLE IX: Ablation study to evaluate the contribution of the physics-assisted polarimetric block on UAVSAR QPOSD.

Model Variant	**mIoU**	**IoU$_0$**	**IoU$_1$**	**IoU$_2$**	**IoU$_3$**	**F1-score**	κ
U-Net (base model)	0.8497	0.9714	0.8425	0.6627	0.9220	0.9100	0.9034
PG-CPB (phase)	0.8700	0.9611	0.8567	0.7258	0.8963	0.9201	0.9059
PG-CPB (power ratios)	0.8811	0.9664	0.8876	0.7704	0.8998	0.9411	0.9196
PG-CPB (coherence)	0.9006	0.9748	0.9095	0.7900	0.9283	0.9511	0.9350
PG-CPB (all physical features)	0.9469	0.9889	0.9582	0.8747	0.9706	0.9728	0.9696
PG-CPB (all physical features + attention)	**0.9556**	**0.9861**	**0.9708**	**0.8997**	**0.9539**	**0.9755**	**0.9739**
PG-CPB (only attention mechanisms)	0.9366	0.9829	0.9527	0.8600	0.9579	0.9558	0.9606

class suggests that the attention mechanism can better exploit the complementary physical information to distinguish oil spills from regions exhibiting similar backscattering characteristics.

3) Overall Contribution of PG-CPB: The ablation results in Table IX demonstrate that the proposed physics-assisted components provide complementary benefits to the segmentation framework. Compared with the baseline U-Net, each individual polarimetric descriptor group improves the segmentation performance, with phase, scattering power ratios, and coherence yielding mIoU gains of 2.03, 3.14, and 5.09 percentage points, respectively. These results indicate that the extracted polarimetric descriptors provide additional discriminative information beyond the features learned directly from the input representation.

The attention mechanism also provides a substantial independent contribution. The PG-CPB variant using only the attention mechanisms achieves an mIoU of 0.9366, corresponding to an improvement of 8.69 percentage points over the baseline U-Net. When all physical descriptors are incorporated without attention, the mIoU further increases to 0.9469, yielding a gain of 9.72 percentage points. Finally, integrating both the complete physical descriptor set and the proposed physics-assisted attention mechanism produces the best overall performance, with an mIoU of 0.9556, an F1-score of 0.9755, and a Cohen's κ coefficient of 0.9739.

Compared with the baseline U-Net, the complete PG-CPB configuration improves the mIoU by 10.59 percentage points, while the oil-spill IoU increases from 0.8425 to 0.9708 and the look-alike IoU increases from 0.6627 to 0.8997. The particularly large improvement in the look-alike category demonstrates the benefit of combining complementary polarimetric scattering descriptors with adaptive attention, which enhances the discrimination of challenging marine surface phenomena with similar backscattering characteristics.

Table X further summarizes the mIoU improvements of the ablation variants relative to the baseline U-Net. The individual descriptor groups yield mIoU improvements of 2.03, 3.14, and 5.09 percentage-points for phase, scattering power ratios, and coherence, respectively. The attention-only configuration produces a larger improvement of 8.69 percentage points, while incorporating all physical descriptors without attention yields a 9.72 percentage-point improvement. The complete configuration combining all physical descriptors with the proposed attention mechanism achieves the largest gain of 10.59 percentage points. Overall, these results confirm that both the physical descriptors and the attention mechanism contribute positively to segmentation performance, while their combination provides the strongest overall feature representation for marine oil-spill segmentation.

TABLE X: Absolute mIoU improvement of PG-CPB ablation variants over the baseline U-Net.

Model Variant	**Δ mIoU**
PG-CPB (phase)	+0.0203
PG-CPB (power Ratios)	+0.0314
PG-CPB (coherence)	+0.0509
PG-CPB (only attention mechanisms)	+0.0869
PG-CPB (all Physical Features)	+0.0972
PG-CPB (Physical Features + Attention)	+0.1059

IX. Discussion

Modern SAR-based oil-spill detection faces a fundamental tension between statistical learning capacity and physical interpretability. During the past decade, larger neural networks and expanding training dataset have yielded incremental improvements, yet the main challenge, i.e, distinguishing oil slicks from biogenic films or low-wind remains partially solved. This is mostly fue to the fact that conventional DL models tend to learn the statistical distribution of intensities and textures rather than the scattering mechanisms that truly differentiate these surface phenomena.

The proposed CPA-Net framework addresses this representational gap by embedding domain knowledge directly into the learning process. Its main component, the PG-CPB, shifts the paradigm moving from simply providing physics-aware inputs to explicitly constraining the network's attention with physically interpretable cues. Rather than treating polarimetric observables as auxiliary channels that can be ignored, the PG-CPB maps scattering power ratios and complex coherences into channel-wise and spatial attention weights. These weights bias internal feature representations toward pixels exhibiting signatures consistent with oil-induced damping and depolarization, thereby pushing decision logic that remains coherent with the underlying physics. The PG-CPB forces the network to learn from physical features, such as the power ratios and the complex coherences, which are known to be sensitive to the damping effect of mineral oil-spills on capillary waves.

The quantitative results on the QPOSD dataset demonstrate the effectiveness of the proposed CPA-Net model. As reported

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in Table IV, CPA-Net achieves an overall accuracy of $\mathrm{Acc} = 0.9900$, $\mathrm{mIoU} = 0.9556$, $\mathrm{F1} = 0.9755$, and $\kappa = 0.9739$, outperforming all real-valued baselines as well as the complex-valued U-Net while maintaining a comparable number of trainable parameters. The ablation results presented in Table IX further demonstrate that these performance gains are directly associated with the progressive incorporation of physically meaningful polarimetric scattering descriptors. Starting from the baseline U-Net, introducing phase descriptors increases the mIoU from 0.8497 to 0.8700, corresponding to an absolute improvement of 2.03 $\%$. introducing only scattering power ratios descriptors further improves the mIoU to 0.8811, yielding an overall gain of 3.14 $\%$ over the baseline, then only coherence descriptors are introduced to the baseline model, this improves the mToU from 0.8497 to 0.9006 yielding an improvement of 5.09%. When the complete set of physical descriptors is employed, the mIoU increases substantially to 0.9469, representing an improvement of 9.72 $\%$. Finally, integrating the proposed physics-assisted attention mechanism with the complete descriptor set achieves the highest performance, reaching an mIoU of 0.9556, F1-score of 0.9755, and κ of 0.9739, corresponding to an overall improvement of 10.59 $\%$ compared with the baseline model. As summarized in Table X, each ablation variant consistently improves the segmentation accuracy relative to the baseline, demonstrating that the extracted polarimetric descriptors provide complementary discriminative information and that the proposed physics-assisted attention mechanism further enhances feature representation by effectively exploiting these physically meaningful scattering characteristics. Collectively, these results confirm that progressively incorporating polarimetric scattering knowledge enables CPA-Net to more effectively distinguish marine oil spills from challenging look-alike phenomena, leading to consistent improvements across all evaluation metrics.

Architecturally, placing the PG-CPB within the decoder stages of the U-Net is advantageous since the network fuses high-level semantic information from the bottleneck with fine spatial details from the encoder. Injecting physics-assisted attention at this juncture allows the model to refine segmentation maps with informed context: boundaries corresponding to oil–water interfaces are sharpened, and activations in regions whose polarimetric signatures align with clean sea surface or biogenic slicks are suppressed. Qualitative results corroborate this behavior, showing clearer delineation of slick edges and reduced false alarms in visually ambiguous areas.

Overall, CPA-Net demonstrates that combining first-principles scattering parameters with DL can significantly improve both reliability and transparency in SAR oil-spill detection. Future work should evaluate the framework across diverse sensors and environments, explore adaptive weighting of physical cues based on scene context, and integrate physics-constrained segmentation into full operational pipelines that include detection, classification, and false-alarm estimation.

X. Conclusion

This study presented CPA-Net, a physics-assisted deep neural network for marine oil-spill segmentation using quad-polarimetric SAR imagery. By integrating polarimetric scattering information into the feature learning process, CPA-Net bridges the gap between data-driven deep learning and physical interpretation of SAR observations. The proposed framework effectively exploits amplitude, coherence, and phase-related information from the coherency matrix to enhance the separation of oil spills from challenging look-alike regions. Experiments on the UAVSAR QPOSD dataset demonstrate that CPA-Net achieves state-of-the-art segmentation performance, highlighting the value of incorporating domain knowledge into complex-valued neural networks for improved accuracy and interpretability.

The results further show that physically meaningful descriptors, particularly coherence-based features, provide complementary information that strengthens the learned representations. Although the proposed method demonstrates strong performance on UAVSAR L-band data, further validation across different sensors, frequencies, and environmental conditions is required to assess its broader applicability. Future work will focus on improving cross-domain generalization and developing more adaptive physics-assisted models for reliable operational marine oil-spill monitoring.

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